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ON TOWERS OF FUNCTION FIELDS AND THE CONSTRUCTION OF THE CORRESPONDING GOPPA CODES

by

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Abstract

In the field of error-correcting codes, one often wants to work with codes that contain a maximal amount of content and error-correcting data relative to their length. Reed-Solomon codes, which were created in the 1960s, are optimal in this sense.

In the late 1970s, Valeri Goppa introduced *algebraic geometry codes*, or *geometric Goppa codes*, which are a generalization of Reed-Solomon codes. The construction of geometric Goppa codes involves points and functions on algebraic curves. Certain Goppa codes gave the first infinite families of codes whose parameters beat the Glibert-Varshamov bound. With this new link between algebraic geometry and coding theory, mathematicians and computer scientists began to look for towers of curves that give rise to families of codes with a strictly positive percentage of content and error-correcting data. Such towers of curves are called *asymptotically good*.

For a tower to be asymptotically good, the number of points must be proportional to the genus in each level of the tower. Such towers are surprisingly difficult to find. The

existence of asymptotically good towers was known in the 1980s, but it was not until 1995 that explicit examples were found. However, even with these explicit examples, the construction of the corresponding Goppa codes has proved to be difficult.

This dissertation is comprised of four chapters. In the first chapter, I provide preliminaries that combine coding theory, algebraic geometry, and number theory. In the second chapter, I compile a survey of recent papers, outline the methods that reach the limit of current knowledge, and state the open questions. In the third chapter, I study a class of towers of curves for which code construction is relatively straightforward and calculate the genus of each level. Finally, working with a known asymptotically good tower over the finite field with eight elements, I explicitly construct codes from the first three levels and demonstrate a method to construct codes from each subsequent level.

Contents

0 Introduction **1**

0.1 Organization 3

1 Preliminaries **5**

1.1 Reed-Solomon codes 7

1.2 Goppa codes 8

1.3 Towers of codes 11

2 Survey **13**

2.1 Known results 13

2.1.1 Asymptotically optimal towers 14

2.1.2 Asymptotically good towers 19

2.2 Constructing codes 20

2.2.1 Codes from the first few levels of an asymptotically optimal tower . 21

2.2.2 Towards a basis for the second tower of Garcia and Stichtenoth . . . 21

3 Towers arising from C_{ab} curves **23**

3.1 The equations 23

3.2 Valuations 24

3.3 Calculating the genera 27

3.3.1 Valuations of polynomials 28

3.3.2 Valuations of rational functions 30

3.3.3 Gap numbers of monomials 32

3.4 Practicality 36

4 Effective construction **37**

4.1 The tower 37

4.2 Creating codes 41

4.2.1 Level one 42

4.2.2 Level two 43

4.2.3 Level three 45

4.2.4 Level four and beyond 46

4.3 Obtaining plane curves 47

4.3.1 Primitive elements associated to the tower over $\mathbb{F}_8$ 48

4.4 Effective construction 51

4.4.1 Preliminaries 51

4.4.2 Constructing the code for the third level of the tower over $\mathbb{F}_8$ 63

A Function Fields **84**

B Identifying singular points **89**

B.1 Determining singular points on D_i 89

List of Tables

4.1 Behavior of points (x, y) on $C : y^2 + y = x + 1 + 1/x$ with coordinates in $\mathbb{P}(\mathbb{F}_4)$, with $\mathbb{F}_4 \cong \mathbb{F}_2(\rho)$, where $\rho^2 + \rho + 1 = 0$. 41

4.2 Behavior of points (x, y) on $C : y^2 + y = x + 1 + 1/x$ with coordinates in $\mathbb{F}_8 \cong \mathbb{F}_2(\alpha)$, where $\alpha^3 + \alpha^2 + 1 = 0$. 42

4.3 Local equations of $Q_{2,1}$, $Q_{2,3}$, and $Q_{2,4}$ 66

4.4 Local equations of $Q_{2,5}$ 67

4.5 Desingularization of $P_2 = (0 : 1 : 0)$ 68

4.6 Blow-ups of all singular points 69

4.7 Local parametrizations. (Note that the parametrizations at $Q_{2,2}$, $Q_{2,6}$, and $Q_{4,5}$ are conjugates of those at $Q_{2,1}$, $Q_{2,5}$, and $Q_{4,4}$, respectively.) 72

4.8 Divisors associated to certain forms 76

4.9 Pole orders of basis elements for $\mathcal{L}(29Q_{1,1})$ with $x = X/Z$, $y = Y/Z$, and $g_0(x, y) = G_0(x, y, 1) = (x + 1)^4 y(y^7 + 1)$. 79

4.10 Basis elements of $\mathcal{L}(29Q_{1,1})$ evaluated at certain points in $\mathcal{P}$ 83

4.11 Basis elements of $\mathcal{L}(29Q_{1,1})$ evaluated at certain points in $\mathcal{P}$ 83

Chapter 0

Introduction

In order to transmit a message reliably, one would like a system in which the message recipient can be sure that the message received is the message that the sender intended to send. The field of coding theory is based on this idea.

One class of codes commonly used is that of error-detecting codes. Examples of systems using error-correcting codes are ISBN book numbers and credit card numbers. Each ISBN number, for example, contains ten digits. The first nine digits can be any digit from 0 to 9, and they identify a specific book. The tenth digit, which is a "check-digit", is then calculated by a formula involving the first nine digits. The formula is chosen so that if any of the previous nine digits are changed or if two digits are transposed, then the check-digit will be different. The result is that if one was to send an ISBN number that had such an error in it, the recipient could use the check-digit to see that an error has occurred and request the number be sent again.

There are instances in which one cannot request the corrected message be sent again, such as in getting data from a compact disc. It could happen that there is a scratch on the disc, corrupting the data. In this case, one would want an error-correcting code, which is a code where one can not only detect errors, but correct them as well. Reed-Solomon

codes are error-correcting codes that are created with polynomials over finite fields. They are very efficient, both in terms of encoding and transmission, and for these reasons, they are widely used.

In the late 1970s, V. Goppa introduced the *geometric Goppa codes*, which are a generalization of Reed-Solomon codes. Goppa codes provide an interesting link between the fields of algebraic geometry and coding theory. Using some deep algebraic geometry results of the time, it was shown that there exist families of geometric Goppa codes of increasing length that have parameters exceeding the Gilbert-Varshamov bound (which is described in [Sti95], for instance). Prior to the introduction of Goppa codes, no codes were known to exceed the G-V bound. As a result, Goppa codes have attracted a good deal of interest.

The Goppa codes with parameters exceeding the G-V bound come from towers of function fields that have many places of degree one relative to the genus. Much work has been done to find examples of such "asymptotically good" towers. The first explicit examples were given by Garcia and Stichtenoth in [GS95] in 1995 and in [GS96b] in 1996. Since then, other asymptotically good towers have been identified. We will give a brief survey of various results since 1995.

The list of asymptotically good towers is not complete. For instance, to date, there has been no explicit description of an asymptotically good tower over $\mathbb{F}_p$, for p a prime. Thus, there is more work to be done in finding new asymptotically good towers. There is also work to be done in describing the Goppa codes that arise from these towers. In this dissertation, we present work towards both goals.

The main difficulty in deciding whether a given tower of function fields is asymptoti-

cally good is to calculate the genus. We present a large class of towers of function fields arising from equations of C_{ab} curves and give a formula for the genus of each level. These towers have the advantage that, provided one has enough places of degree one, the associated Goppa code is very easy to create. Unfortunately, as we see, these towers are not asymptotically good. In general, it is difficult to create the Goppa code associated to an asymptotically good tower because of the presence of singular points. Work has been done to create codes from the towers of Garcia and Stichtenoth. Here, we will present work towards creating codes from a tower of van der Geer and van der Vlugt from [GV02].

0.1 Organization

This dissertation is organized as follows. In Chapter 1, we define linear codes and discuss their properties. We present the construction of Reed-Solomon and geometric Goppa codes, giving bounds on the parameters of these codes. Then, we give motivation for asymptotically good towers of function fields which lead to asymptotically good codes.

In Chapter 2, we present a survey of results in algebraic geometry coding theory. We begin with the Hasse-Weil bound, which relates the number of points on a curve over a finite field to the genus of the curve, and the Drinfeld-Vlăduţ bound, which gives a similar result for the case when the genus is large. We then present the equations for some asymptotically good towers found in the past ten years. Finally, we present work that has been done towards creating codes from these asymptotically good towers.

In Chapter 3, we first define C_{ab} curves, which are generalizations of elliptic curves in that there is a unique point at infinity P_∞ and there are functions x and y that have poles

of orders b and a at P_∞ (and nowhere else). A reason to study such curves is that the Riemann-Roch space associated to rP_∞ is easy to calculate when the curve is non-singular. We create a tower of function fields from the equation of a C_{ab} curve and calculate the genus of each level of the tower, as well as provide a basis for $\mathcal{L}(rP_\infty^n)$, where P_∞^n is the unique place at infinity in the nth level of the tower.

In Chapter 4, we use an asymptotically good tower over $\mathbb{F}_8$ to create codes. As we see, when one works with higher levels in the tower, one runs into singularities which make the Riemann-Roch space harder to calculate. We explore a deterministic method by finding appropriate plane curves and then using the algorithms of Haché and Le Brigand to construct the corresponding code.

Finally, in Appendix A, we present basic facts about algebraic function fields that are used throughout the dissertation, and in Appendix B, we show the calculations involved in determining the singular points from the tower of van der Geer and van der Vlugt.

Chapter 1

Preliminaries

Throughout this paper, we let $\mathbb{F}_q$ denote the finite field with q elements. Given two vectors $u, v \in \mathbb{F}_q^n$ with

$$u = (u_1, u_2, \ldots, u_n) \text{ and } v = (v_1, v_2, \ldots, v_n),$$

we define the distance to be $d(u, v) = \#\{i \mid u_i \neq v_i\}$. This distance function is a metric on $\mathbb{F}_q^n$. In particular, it satisfies the triangle inequality

$$d(u, v) + d(v, w) \geq d(u, w)$$

for all $u, v, w \in \mathbb{F}_q^n$. The weight of u, denoted $wt(u)$ is the number of non-zero elements of u, i.e., $wt(u) = d(u, 0)$.

We define a *code* C of length n over $\mathbb{F}_q$ to be a linear subspace of $\mathbb{F}_q^n$. Elements of C are *codewords*. There are non-linear codes, but we will not discuss them here.

Let k be the dimension of C over $\mathbb{F}_q$ (so $\#C = q^k$). The minimum distance $d = d(C)$ is the minimum distance between any two distinct codewords, i.e.,

$$d(C) = \min\{d(a, b) \mid a, b \in C, a \neq b\}.$$

Since C is a linear code, one also has

$$d(C) = \min\{wt(u) : u \in C, u \neq 0\}.$$

We will refer to such a code C as a $[n, k, d]_q$ code. (We omit the q when it is not necessary.)

An $[n, k, d]_q$ code can correct up to $\left\lfloor \frac{d-1}{2} \right\rfloor$ errors (where $\lfloor x \rfloor$ denotes the integer part of x, i.e. $x = \lfloor x \rfloor + \epsilon$ with $\lfloor x \rfloor \in \mathbb{Z}$ and $0 \leq \epsilon < 1$). For any vector $a \in \mathbb{F}_q^n$, there is at most one codeword $c \in C$ with $d(a, c) \leq \left\lfloor \frac{d-1}{2} \right\rfloor$. If there were two, say c_1 and c_2, then, since $d(u, v)$ satisfies the triangle inequality, we have

$$d(c_1, c_2) \leq d(c_1, a) + d(a, c_2) \leq d - 1.$$

Hence, $d(c_1, c_2) = 0$, so $c_1 = c_2$.

In general, one wants both k and d to be large with respect to n: larger k means there are more codewords to work with, and larger d means one can tolerate more errors. The first bound one encounters is the Singleton bound, which says

$$k + d \leq n + 1$$

for any $[n, k, d]_q$ code C.

To see this, given an $[n, k, d]_q$ code C, consider the subspace $U \subset \mathbb{F}_q^n$ given by

$$\begin{aligned} U &= \{u = (u_1, u_2, \ldots, u_n) : u_i = 0, \forall i \geq d\} \\ &= \{u = (u_1, \ldots, u_{d-1}, 0, \ldots, 0)\}. \end{aligned}$$

The dimension of U is $d-1$. Every $u \in U$ has at most $d-1$ non-zero entries, meaning that $wt(u) \leq d-1$ for all $u \in U$. Since the minimum distance of C is d, we have that $U \cap C = \{0\}$, so $\dim U + \dim(C) \leq n$. Thus,

$$d - 1 + k \leq n.$$

There exist more restrictive bounds when which also account for the size of q. References can be found in [Sti95] on page 140.

1.1 Reed-Solomon codes

A class of codes commonly used in practice today is that of Reed-Solomon codes. To create a Reed-Solomon code over $\mathbb{F}_q$, one needs to choose distinct elements $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q$ (so $n \leq q$). For any $r \in \mathbb{N}$, consider the vector space

$$\mathcal{L}_r = \{f(x) \in \mathbb{F}_q[x] : \deg f \leq r\},$$

along with the evaluation map $ev : \mathcal{L}_r \to \mathbb{F}_q^n$ given by

$$ev(f) = (f(\alpha_1), \ldots, f(\alpha_n)).$$

The Reed-Solomon code C is then

$$C = \{ev(f) : f \in \mathcal{L}_r\}.$$

What are the parameters of this code? The length of the code is n. The kernel of the map ev is

$$\{f \in \mathcal{L}_r : f(\alpha_1) = f(\alpha_2) = \cdots = f(\alpha_n) = 0\}.$$

Since $\mathbb{F}_q$ is a field, a polynomial of degree r has at most r roots. We suppose $r < n$ because in this case, the kernel of ev is $\{0\}$, so ev is an injection. The dimension of $\mathcal{L}_r$ is $r+1$ (a basis is $\{1, x, x^2, \ldots, x^r\}$), so the dimension k of C is also $r+1$. Finally, as there can be at most r zeroes in a non-zero codeword, the distance of the code is at least $n-r$. By the Singleton bound,

$$n+1 \geq d+k \geq (n-r)+(r+1) = n+1.$$

Since $k = r+1$, we must have $d = n-r$.

To summarize, C is a $[n, r+1, n-r]_q$ code. Reed-Solomon codes are optimal in the sense that they attain the Singleton bound. However, one is restricted to the case where $n \leq q$, so in order to work with longer codewords, one has to work in a larger field.

1.2 Goppa codes

In the late 1970s, V. Goppa generalized the Reed-Solomon codes using algebraic geometry. The construction involves algebraic function fields. There is a brief description of properties of algebraic function fields given in Appendix A.

We can describe them as follows. Consider an algebraic function field $F/\mathbb{F}_q$ of genus g. Fix a finite set $\mathcal{P} = \{P_1, \ldots, P_n\}$ of places of $F/\mathbb{F}_q$ of degree one and a divisor D such that $\mathcal{P}$ and the support of D do not intersect. For the Riemann-Roch space

$$\mathcal{L}(D) = \{f \in F : (f) + D \geq 0\} \cup \{0\},$$

consider the map

$$ev : \mathcal{L}(D) \to \mathbb{F}_q^n$$

where

$$ev(f) = (f(P_1), f(P_2), \ldots, f(P_n)).$$

The code

$$C(\mathcal{P}, D) = \{ev(f) : f \in \mathcal{L}(D)\} \subseteq \mathbb{F}_q^n$$

is called the *algebraic geometric code* associated to the divisor D and the places $P_1, \ldots, P_n$. Since any function $f \in \mathcal{L}(D)$ can only have poles at places in D, and since the places P_i are places of degree one, $f(P_i) \in \mathbb{F}_q$ is well-defined for $i = 1, \ldots, n$.

Given such a code, we can find its parameters. The length of the code is n. A function $f \in \mathcal{L}(D)$ has at most $\deg D$ poles, and thus at most $\deg D$ zeroes, so if $\deg D < n$, the evaluation map is injective, and so the dimension of $C(\mathcal{P}, D)$ is the dimension of $\mathcal{L}(D)$. By the Riemann-Roch Theorem (stated in Appendix A), the dimension k of $\mathcal{L}(D)$ is at least $\deg D + 1 - g$. Furthermore, if $\deg D > 2g - 2$, then $k = \deg D + 1 - g$.

To find the minimum distance d, suppose we have a function $f \in \mathcal{L}(D)$ such that

$ev(f)$ has a maximal number of zeroes (suppose $f(P_1) = \cdots = f(P_{n-d}) = 0$). Then $f \in \mathcal{L}(D-(P_1+\cdots+P_{n-d}))$. This means $\mathcal{L}(D-(P_1+\cdots+P_{n-d})) \neq \emptyset$, so $\deg D-(n-d) \geq 0$. We call $d^* = n - \deg D$ the *designed distance* and have $d \geq d^*$.

We summarize these results to get the *Goppa bound*. Since $k \geq \deg D + 1 - g$ and $d \geq n - \deg D$, we get that $k + d \geq n + 1 - g$. Combined with the Singleton bound, we thus have

$$n + 1 - g \leq k + d \leq n + 1.$$

Note that we require $\deg D < n$.

Example 1. *Consider the rational function field $F = K(x)$ for $K = \mathbb{F}_q$. There are $q + 1$ places of degree one in F: q of them correspond to the polynomials $x - \alpha_i$ for $\alpha_i \in K$, and one corresponds to the polynomial $1/x$. Let P_i be the place corresponding to $x - \alpha_i$ and P_∞ be the place corresponding to $1/x$, i.e.*

$$P_i = \left\{ \frac{(x - \alpha_i)f(x)}{g(x)} : f(x), g(x) \in K[x], (x - \alpha) \nmid g(x) \right\}$$

and

$$P_\infty = \left\{ \frac{f(x)}{g(x)} : \deg(g) > \deg(f) \right\}.$$

Let $D = rP_\infty$ and

$$\mathcal{P} = \{P_1, \ldots, P_q\}.$$

A basis for $\mathcal{L}(D)$ is

$$\left\{1, x, x^2, \ldots, x^r\right\},$$

and for $f \in \mathcal{L}(D)$, we have $f(P_i) = f(\alpha_i)$. Then the code $C(\mathcal{P}, D)$ is a Reed-Solomon code, as described in Section 1.1.

As we see in this example, Goppa codes are a generalization of Reed-Solomon codes, with the advantage that we are no longer restricted to codes of length $n \leq q$. One can construct arbitrarily long codes while working over $\mathbb{F}_q$. However, as we will see, these codes are more difficult to construct than Reed-Solomon codes because, in general, it is difficult to calculate a basis for $\mathcal{L}(D)$.

1.3 Towers of codes

Now that we have codes of arbitrary length, we need a way to compare their parameters to classify good codes. Given an $[n, k, d]$ code, we define the *rate* $R := k/n$ and the *relative minimum distance* $\delta := d/n$. Just as one wants k and d to be large, one wants R and δ to be large.

From the Goppa and Singleton bounds, we have

$$n + 1 - g \leq k + d \leq n + 1$$

so

$$1 + \frac{1}{n} - \frac{g}{n} \leq R + \delta \leq 1 + \frac{1}{n}.$$

We now have a lower bound on what parameters algebraic geometric codes can have. If we minimize $\frac{g}{n}$, we maximize our lower bound.

We can consider a tower $\mathcal{F}$ of function fields $F_i/\mathbb{F}_q$, $F_i \subsetneq F_{i+1}$, having genus g_i and n_i places of degree one for $i = 1, 2, \ldots$. Suppose $g_i \to \infty$ as $i \to \infty$ and

$$\lim_{i \to \infty} \frac{n_i}{g_i} = \gamma > 1.$$

Then, since $n_i \to \infty$, our bound becomes

$$R + \delta \geq 1 - \gamma^{-1}.$$

To simplify notation, we let

$$\lambda(\mathcal{F}) = \lim_{i \to \infty} \frac{n_i}{g_i}.$$

It is shown in [GS96b] that this sequence converges for any tower $\mathcal{F}$.

In order to get an explicit non-zero lower bound for $R + \delta$, we are interested in towers $\mathcal{F}$ such that $\lambda(\mathcal{F}) > 1$. In fact, we will look for towers with $\lambda(\mathcal{F}) > 0$. Such towers are said to be *asymptotically good*.

Chapter 2

Survey

2.1 Known results

For a function field $F/\mathbb{F}_q$, let $n = n(F)$ be the number of places of degree one of F and $g = g(F)$ the genus of F. The Hasse-Weil bound (as presented in [Sti93], for instance) states that

$$n \leq q + 1 + 2g\sqrt{q}.$$

This bound is known to be sharp when the genus is small relative to q.

Let

$$n_q(g) = \max\{n(F) : F \text{ is a function field over } \mathbb{F}_q \text{ of genus } g\},$$

and let

$$A(q) = \limsup_{g\to\infty} n_q(g)/g.$$

By the Hasse-Weil bound, we have

$$A(q) \leq 2\sqrt{q}.$$

In [DV83], Drinfeld and Vlăduţ improved the bound, proving that

$$A(q) \leq \sqrt{q} - 1.$$

If q is a square, then this bound is the best possible. If q is not a square, less is known.

One fact, which is quite clear, is that for any tower $\mathcal{F}$ over $\mathbb{F}_q$, we have $0 \leq \lambda(\mathcal{F}) \leq A(q)$. Towers $\mathcal{F}$ for which $\lambda(\mathcal{F}) = A(q)$ are called *asymptotically optimal*.

2.1.1 Asymptotically optimal towers

For fields of cardinality q^2, it was shown independently by Ihara and Tsfasman, Vlăduţ, and Zink (described in [GS95], [Elk97], [Ser83], [Sti95], and [Deo00]) that there exist towers $\mathcal{F}$ over $\mathbb{F}_{q^2}$ such that

$$\lambda(\mathcal{F}) = q - 1.$$

Thus, we have

$$A(q^2) = q - 1.$$

They used modular curves to prove the existence of these towers without giving explicit equations. On pages 178–211 in [Mor91], the author gives good background information leading up to the Tsfasman-Vlăduţ-Zink towers.

In [GS95], Garcia and Stichtenoth gave explicit equations for a tower attaining the Drinfeld-Vlăduţ bound. The tower $\mathcal{F} = (F_1, F_2, F_3, \dots)$ is given as follows. Let $F_1 = \mathbb{F}_{q^2}(x_1)$ be the rational function field over $\mathbb{F}_{q^2}$. For $n \geq 1$, let

$$F_{n+1} = F_n(z_{n+1}),$$

where

$$z_{n+1}^q + z_{n+1} = x_n^{q+1}$$

with

$$x_n = z_n / x_{n-1} \in F_n \text{ (for } n \geq 2).$$

They were able to understand the ramification behavior, and thus calculate the genus and number of places, because these are generalized Artin-Schreier extensions.[1] Their result is that

$$\lambda(\mathcal{F}) = q - 1,$$

so the tower is asymptotically optimal. The fact that $\lambda(\mathcal{F}) = q - 1$, along with the Drinfeld-Vlăduţ bound, gives another proof that

$$A(q^2) = q - 1.$$

In a subsequent paper [GS96b], they considered the tower $\mathcal{T} = (T_1, T_2, T_3, \dots)$ of function fields over $\mathbb{F}_{q^2}$ given by $T_n = \mathbb{F}_{q^2}(x_1, \dots, x_n)$ where

$$x_{i+1}^q + x_{i+1} = \frac{x_i^q}{x_i^{q-1} + 1},$$

for $i = 1, \dots, n$. Again, these are generalized Artin-Schreier extensions, and they found that

$$\lambda(\mathcal{T}) = q - 1,$$

[1] Artin-Schreier extensions are defined and their properties are discussed in Section 4.1.

so $\mathcal{T}$ is also asymptotically optimal. In fact, it turns out that $\mathcal{T}$ is a subtower of the tower $\mathcal{F}$ given above.

These towers of Garcia and Stichtenoth are summarized in [GS96a]. In this paper, the authors also mention two other towers which are asymptotically optimal in certain cases. These towers involve Kummer extensions with tame ramification. The details of their calculations for the towers of Kummer extensions can be found in [GST97]. We present the results below.

For the first tower $\mathcal{F} = (F_1, F_2, \dots)$, let $q = p^e$ for $e > 1$, and set $m = (q-1)/(p-1)$. For $n > 1$, F_n is defined by $F_n = \mathbb{F}_q(x_1, \dots, x_n)$ where

$$x_{i+1}^m + (x_i + 1)^m = 1$$

for $i = 1, \dots, n-1$. The result is that

$$\lambda(\mathcal{F}) \geq \frac{2}{q-2}.$$

Notice that for $q = 4$, $\lambda(\mathcal{F}) \geq 1 = \sqrt{4} - 1$. Thus, this tower is optimal for $q = 4$. Further notice that this proves $A(q) > 0$ for q not a prime number.

For the second tower, assume that $q = l^2 > 4$. For $n > 1$, F_n is defined by $F_n = F_q(x_1, \dots, x_n)$ where

$$x_{i+1}^{l-1} + (x_i + 1)^{l-1} = 1$$

for $i = 1, \dots, n-1$. Here,

$$\lambda(\mathcal{F}) \geq \frac{2}{l-2} = \frac{2}{\sqrt{q}-2}.$$

For $q = 9$, this tower is optimal.

Towers of modular curves

In [Elk97], Elkies looked at towers of classical modular curves $\{X_0(l^n)\}_{n>1}$. For small values of l, it is possible to find explicit equations for these curves.

For $l = 2$, one finds that $X_0(2^n)$ can be defined by the $n-1$ equations

$$(x_j^2 - 1)(z_{j+1}^2 - 1) = 1 \qquad (j = 1, \ldots, n-2),$$

where

$$z_j = (x_j + 3)/(x_j - 1).$$

With the substitution $y_j = 1 - x_j^{-1}$ in characteristic 3, this produces

$$y_{j+1}^2 = y_j - y_j^2,$$

which is the second tower of Kummer extensions of Garcia and Stichtenoth from the previous section with $q = 9$.

For $l = 3$, one finds $X_0(3^n)$ can be defined by

$$(x_j^3 - 1)(z_{j+1}^3 - 1) = 1 \qquad (j = 1, \ldots, n-2),$$

where

$$z_j = (x_j + 2)/(x_j - 1).$$

With the same substitution in characteristic 2, one has

$$y_{j+1}^3 = y_j^3 + y_j^2 + y_j.$$

This is the first tower of Kummer extensions of Garcia and Stichtenoth in the previous section with $q = 4$.

In [Elk01], Elkies found explicit equations for towers of Drinfeld modular curves over finite fields. These equations turn out to be the equations for the asymptotically optimal towers of Artin-Schreier extensions found by Garcia and Stichtenoth in [GS95] and [GS96b] (which are described here in Section 2.1.1).

Elkies speculates in [Elk97] that perhaps all asymptotically optimal towers come from modular curves.

Symmetric functions

Deolalikar used the idea of symmetric functions and quasi-symmetric functions[2] in [Deo00] to construct towers $\mathcal{F}$ in which all, or all except one, rational places split completely throughout the tower. The purpose of this is to maximize the number of places in every step of the tower. Unfortunately, the ramification in the non-rational places rises quickly, which in turn makes the genus rise quickly. For these towers, $\lambda(\mathcal{F}) = 0$. As a result, Deolalikar remarks that one may need a certain amount of ramification in the rational places in order to attain towers $\mathcal{F}$ with $\lambda(\mathcal{F}) > 0$.

In addition to finding towers with all rational places splitting completely, Deolalikar

[2] A polynomial $f(x_1, x_2, \ldots, x_n)$ is *symmetric* if it is fixed under the action of the permutation group S_n on n elements. f is *quasi-symmetric* if it is fixed by the permutation $(12\ldots n) \in S_n$.

also generalized the two Kummer extension towers of Garcia and Stichtenoth which were described above.

2.1.2 Asymptotically good towers

For q not a square, it is not known if the upper bound of $A(q) \leq \sqrt{q}-1$ is sharp. In [Ser83], Serre mentions that from class-field theory, one has $A(q) \geq c \cdot \log q$ for some constant $c > 0$. Lower bounds have been found for small values of q.[3]

Zink (as mentioned in [GV02] and [BGS]) used Shimura modular surfaces in a tower $\mathcal{C}$ over $\mathbb{F}_{p^3}$, for p a prime, such that

$$\lambda(\mathcal{C}) \geq \frac{2(p^2-1)}{p+2},$$

but the construction is not explicit.

In [GV02], van der Geer and van der Vlugt found an explicit tower $\mathcal{C}$ of Artin-Schreier extensions over $\mathbb{F}_8$ with the recursive equation

$$x_{i+1}^2 + x_{i+1} = x_i + 1 + 1/x_i.$$

They found that

$$\lambda(\mathcal{C}) = 3/2.$$

The Drinfeld-Vlăduţ bound gives us that $A(8) \leq \sqrt{8} - 1 \approx 1.83$, so this is not far from optimal. The basic idea of what they did can be found in Section 4.1.

[3] Some of these values, as well as references, can be found in [GS96b].

In [BGS], the authors generalized the tower of van der Geer and van der Vlugt to find asymptotically good towers over all fields of cubic cardinality, extending Zink's result. For their tower $\mathcal{F}/\mathbb{F}_{q^3}$, they used the recursive equation

$$\frac{1 - x_{i+1}}{x_{i+1}^q} = \frac{x_i^q + x_i - 1}{x_i}.$$

Such a tower has $\lambda(\mathcal{F}) = \frac{2(q^2-1)}{q+2}$. When $q = 2$, this is essentially the tower of van der Geer and van der Vlugt. However, for $q > 2$, the extensions are no longer Artin-Schreier extensions - they're not even Galois - so the work required to calculate the genus is quite technical.

2.2 Constructing codes

The motivation for finding these asymptotically good towers of curves comes from the resulting codes. The difficulty in creating the codes coming from the curves lies in finding a basis for the Riemann-Roch space $\mathcal{L}(D)$ for some divisor D. We will highlight two approaches to creating codes from the asymptotically optimal towers of Garcia and Stichtenoth. In the first subsection, the authors try to find a basis for $\mathcal{L}(D)$ using monomials, and the result is that they do not have enough functions. In the following subsection, the authors look at a slightly larger class of functions that spans $\mathcal{L}(D)$ but also includes functions outside of $\mathcal{L}(D)$.

2.2.1 Codes from the first few levels of an asymptotically optimal tower

In [VH97], the authors used the first few levels of Garcia and Stichtenoth's first tower (from [GS95], as described in Section 2.1.1) to construct explicit codes over every field $\mathbb{F}_{q^2}$. For the first two levels, finding a basis for the Riemann-Roch space for a certain divisor (and hence, the construction of the codes) is straightforward - the first level is a rational function field, and the second level is a Hermitian function field. In the third level, the curve has singularities, so the Riemann-Roch space is harder to identify. However, using certain monomials, they are able to generate functions with every required pole order, so they have an explicit description of the Riemann-Roch space for the third level.

They go on to comment that for the fourth level (and beyond), a similar space of monomials will not span the Riemann-Roch space. They looked at the codes coming from these subspaces of monomials and concluded that these codes are unlikely to be asymptotically optimal. Thus, to create an asymptotically good tower of codes, one needs to consider a wider class of function in the Riemann-Roch space.

2.2.2 Towards a basis for the second tower of Garcia and Stichtenoth

Working with the second tower of Garcia-Stichtenoth (from [GS96b], described in Section 2.1.1) the authors of [ADKS01] present work towards finding $\mathcal{L}(rP)$, for $r \in \mathbb{N}$, the extension T_n/T_1, and a place P in T_n. We summarize their results here.

Let $g_1 = x_1^{q-1} + 1$ and

$$\mathfrak{G}_{g_1} = \{1\} \cup \left\{ g_1 \prod_{i=2}^{n} x_i^{e_i} : 0 \leq e_i \leq q-1, \text{ at least one } e_i \neq 0 \right\}.$$

Let P_0 and P_∞ be the unique zero and pole (respectively) of x_1. Let $\mathbb{P}_{T_i}$ denote the places of T_i, $S = \mathbb{P}_{T_1} - \{P_0, P_\infty\}$, and $S^{T_n} = \{P' \in \mathbb{P}_{T_n} : P'|P \text{ for } P \in S\}$. Then there is the following theorem:

Theorem 1. *Let $f \in T_n$. If $v_{P'}(f) \geq 0$ for every place P' of S^{T_n} (i.e., if f has no poles at any place P' in S^{T_n}), then f can be written as a linear combination of functions in the set $\mathfrak{G}_{g_1}$ with coefficients of the form $p(x_1)/x_1^i$ where $p(x_1)$ is a polynomial in x_1.*

Every function $f \in \mathcal{L}(rP)$ for $P = P_0$ or P_∞ has the above property. Unfortunately, this theorem doesn't say anything about the poles at P_0 or P_∞, so there are functions that satisfy the hypothesis of the theorem but aren't in $\mathcal{L}(rP)$.

Chapter 3

Towers arising from C_{ab} curves

3.1 The equations

Let q be a prime power, $\mathbb{F}_q$ be the finite field with q elements, and $a > b$ positive integers with $\gcd(a, b) = 1$. Consider the function $f(x, y)$ of two variables given by

$$f(x, y) = \alpha_{a,0}x^a + \alpha_{0,b}y^b + \sum_{i,j} \alpha_{i,j}x^i y^j,$$

where the sum is over i and j non-negative integers such that $aj + bi < ab$. A curve $C : f(x, y) = 0$ is called a C_{ab} curve. These curves are known to be absolutely irreducible (see [Mat98], for instance).

Let

$$C_i = \left\{(p_0, p_1, \ldots, p_i) \in \mathbb{P}^1 \times \cdots \times \mathbb{P}^1 : (p_{j-1}, p_j) \in C \text{ for } j = 1, \ldots, i\right\}.$$

We have an associated tower of function fields $\mathcal{F} = (F_0, F_1, \ldots)$, where $F_0 = \mathbb{F}_q(x_0)$ and $F_n = F_{n-1}(x_n)$ for $n \geq 1$. The x_k are related by the equation

$$f(x_{k-1}, x_k) = 0$$

for $k = 1, \ldots, n$.

C_{ab} curves are nice due to the fact that the poles of x and y are precisely bP_∞ and aP_∞ (respectively) for P_∞ the unique place at infinity. As we will see in the following section, one can easily determine the poles of x_k for any k. We will use this information to determine what pole orders will appear in the Riemann-Roch space of a specific divisor, which lets us calculate the genus of the function field.

3.2 Valuations

F_0 is a rational function field. The divisor associated to the function x_0 is $P_0 - P_\infty$, where

$$P_0 = \left\{ \frac{x_0 \cdot f(x_0)}{g(x_0)} : f(x_0), g(x_0) \in \mathbb{F}_q[x_0], x_0 \nmid g(x_0) \right\}$$

and

$$P_\infty = \left\{ \frac{f(x_0)}{g(x_0)} : \deg g > \deg f \right\}.$$

To each place, we have an associated valuation. For the purposes of this paper, we are only interested in the valuation associated to P_∞, which is defined as

$$v_\infty \left(\frac{f(x)}{g(x)} \right) = \deg g - \deg f.$$

Proposition 2. *Let P_∞^n denote a place in $\mathbb{P}_{F_n}$ lying above P_∞, and let v_∞^n be the valuation associated to that place. Then P_∞^n is the only place in $\mathbb{P}_{F_n}$ above P_∞ and $v_\infty^n(x_n) = -a^n$.*

Proof. For induction, suppose $n = 0$. The statement holds because we have $F_0 = \mathbb{F}_q(x_0)$, so P_∞^0 is the unique place at infinity in the rational function field and $v_\infty^0(x_0) = -1$.

Now, suppose the statement is true for $n = k$. Since x_k and x_{k+1} satisfy the equation

$$\alpha_{a,0}x_k^a + \alpha_{0,b}x_{k+1}^b + \sum_{aj+bi<ab} \alpha_{i,j}x_k^i x_{k+1}^j = 0,$$

we have

$$\begin{aligned} av_\infty^{k+1}(x_k) &= v_\infty^{k+1}\left(\alpha_{a,0}x_k^a\right) \\ &= v_\infty^{k+1}\left(\alpha_{0,b}x_{k+1}^b + \sum_{aj+ib<ab} \alpha_{i,j}x_k^i x_{k+1}^j\right) \\ &\geq \min\left(\{bv_\infty^{k+1}(x_{k+1})\} \cup \{iv_\infty^{k+1}(x_k) + jv_\infty^{k+1}(x_{k+1}) : aj + ib < ab\}\right), \end{aligned}$$

by the triangle inequality. For the sake of contradiction, suppose

$$iv_\infty^{k+1}(x_k) + jv_\infty^{k+1}(x_{k+1}) \leq bv_\infty^{k+1}(x_{k+1})$$

for some non-negative i, j with $aj + bi < ab$ such that the left-hand side of the inequality is minimal. Then

$$\frac{v_\infty^{k+1}(x_k)}{v_\infty^{k+1}(x_{k+1})} \geq \frac{b-j}{i},$$

and since $aj + bi < ab$, we have $\frac{b-j}{i} > \frac{b}{a}$, so

$$\frac{v_\infty^{k+1}(x_k)}{v_\infty^{k+1}(x_{k+1})} > \frac{b}{a},$$

which means

$$\frac{a}{b} > \frac{v_\infty^{k+1}(x_{k+1})}{v_\infty^{k+1}(x_k)}.$$

On the other hand, if

$$iv_\infty^{k+1}(x_k) + jv_\infty^{k+1}(x_{k+1}) \leq bv_\infty^{k+1}(x_{k+1})$$

for i and j with $aj + bi < ab$ such that the left-hand side is minimal, then by the triangle inequality above,

$$av_\infty^{k+1}(x_k) \geq iv_\infty^{k+1}(x_k) + jv_\infty^{k+1}(x_{k+1}).$$

This implies

$$\frac{a-i}{j} \leq \frac{v_\infty^{k+1}(x_{k+1})}{v_\infty^{k+1}(x_k)}.$$

Since $aj + bi < ab$, we have $\frac{a}{b} < \frac{a-i}{j}$. So

$$\frac{a}{b} < \frac{v_\infty^{k+1}(x_{k+1})}{v_\infty^{k+1}(x_k)}.$$

We have a contradiction to our initial assumption. Therefore, we must have

$$bv_\infty^{k+1}(x_{k+1}) < iv_\infty^{k+1}(x_k) + jv_\infty^{k+1}(x_{k+1})$$

for all i, j with $aj + bi < ab$. Using the strict triangle inequality, we see that

$$av_\infty^{k+1}(x_k) = bv_\infty^{k+1}(x_{k+1})$$

.

Since $\gcd(a, b) = 1$, b must divide $v_\infty^{k+1}(x_k)$. Let e denote the ramification degree of P_∞^{k+1} over P_∞^k, so

$$e = \frac{v_\infty^{k+1}(f)}{v_\infty^k(f)}$$

for any $f \in P_\infty^k$. Taking $f = x_k$, we have $ev_\infty^k(x_k) = v_\infty^{k+1}(x_k)$. By the inductive hypothesis, $v_\infty^k(x_k) = -a^k$, so b must divide e. However, since F_{k+1}/F_k is an extension of degree b, the ramification degree is at most b. Therefore, $e = b$, so P_∞^{k+1} is totally ramified over P_∞^k. Since ramification behaves well in towers, P_∞^{k+1} is totally ramified over P_∞^0, and thus unique.

From the formula for the ramification index,

$$\begin{aligned} v_\infty^{k+1}(x_k) &= e v_\infty^k(x_k) \\ &= b(-a^k). \end{aligned}$$

Then, since $a v_\infty^{k+1}(x_k) = b v_\infty^{k+1}(x_{k+1})$, we have

$$v_\infty^{k+1}(x_{k+1}) = -a^{k+1},$$

as desired.

By induction, the statement is true for all non-negative integers n. □

Corollary 3. *For $n \geq k$,*

$$v_\infty^n(x_k) = -a^k b^{n-k}.$$

Proof. By the formula for the ramification index, we have

$$e(P_\infty^n | P_\infty^k) = \frac{v_\infty^n(x_k)}{v_\infty^k(x_k)},$$

so

$$v_\infty^n(x_k) = -a^k b^{n-k}.$$

From the previous proposition, we know P_∞^n is totally ramified over P_∞^k for all $k < n$ because ramification works transitively in towers. Thus, $e(P_\infty^n | P_\infty^k) = [F_n : F_k] = b^{n-k}$. We also know $v_\infty^k(x_k) = -a^k$. The result follows immediately. □

3.3 Calculating the genera

In order to calculate the genera, we will count the number of Weierstrass gap numbers (which are described in Appendix A) of P_∞^n in $F_n/\mathbb{F}_q$. We do so by considering the

pole orders of monomials. In the first subsection, we show that the set of pole orders of polynomials is the same as that of monomials. In the second subsection, we show that, under certain conditions, one obtains no new pole orders with rational functions. In the third subsection, we calculate the number of Weierstrass gap numbers resulting from monomials, which gives us the genus of the function field.

3.3.1 Valuations of polynomials

Let

$$I_n = (f(x_0, x_1), \ldots, f(x_{n-1}, x_n)) \subset F_n$$

be the ideal of the curve C_n, and let

$$\Gamma = \mathbb{F}_q[x_0, \ldots, x_n]/I_n$$

be the coordinate ring of C_n. For notation, let $R_b = \{0, 1, \ldots, b-1\}$ be the set of residues mod b.

Claim 1. *Any polynomial $g(x_0, \ldots, x_n) \in \Gamma$ can be written as*

$$g(x_0, \ldots, x_n) = \sum_{\mathbf{e} \in \mathbb{Z}^{\geq 0} \times (R_b)^n} \lambda_{\mathbf{e}} x_0^{e_0} \ldots x_n^{e_n},$$

for $\mathbf{e} = (e_0, e_1, \ldots, e_n)$ and $\lambda_{\mathbf{e}} \in \mathbb{F}_q$. In particular, for $i = 1, \ldots, n$, one has $0 \leq e_i < b$.

Proof. Given the polynomial g, one can first reduce all powers of x_n to be less than b using the relation $f(x_{n-1}, x_n) = 0$. One can then reduce all powers of x_{n-1} to be less than b using the relation $f(x_{n-2}, x_{n-1}) = 0$. As this does not affect powers of x_n, one can continue on to reduce all powers of x_{n-2}, ..., x_1, giving the resulting form. $\square$

We will call a polynomial written in this form *b-reduced*.

Claim 2. *Let* $g(x_0, \ldots, x_n) = x_0^{e_0} \ldots x_n^{e_n}$ *and* $h(x_0, \ldots, x_n) = x_0^{d_0} \ldots x_n^{d_n}$ *be two monomials in* Γ *with* $0 \leq e_i, d_i < b$ *for* $i = 1, \ldots, n$. *Then* $v_\infty^n(g) = v_\infty^n(h) \iff g = h$.

Proof. Since the pole order of x_i is $a^i b^{n-i}$, this means that

$$\sum_{i=0}^{n} e_i a^i b^{n-i} = \sum_{j=0}^{n} d_j a^j b^{n-j}.$$

Reducing modulo b,

$$e_n a^n \equiv d_n a^n \mod b.$$

Since $\gcd(a^n, b) = 1$ and $0 \leq e_n, d_n < b$, we have $e_n = d_n$. So

$$\sum_{i=0}^{n-1} e_i a^i b^{n-i} = \sum_{j=0}^{n-1} d_j a^j b^{n-j}.$$

Dividing through by b, we similarly obtain $e_{n-1} = d_{n-1}$, and so on to $e_1 = d_1$. Thus, $e_0 = d_0$, and so $g = h$. □

Claim 3. *Let* $g(x_0, \ldots, x_n) \in \Gamma$ *be a polynomial. Then there exist constants* $\lambda_{\mathbf{e}} \in \mathbb{F}_q$ *such that*

$$g(x_0, \ldots, x_n) = \sum_{\mathbf{e} \in \mathbb{Z}^{\geq 0} \times (R_b)^n} \lambda_{\mathbf{e}} x_0^{e_0} \ldots x_n^{e_n},$$

and

$$v_\infty^n(g) = \min \left\{ v_\infty^n(x_0^{e_0} \ldots x_n^{e_n}) : \lambda_{\mathbf{e}} \neq 0 \right\}.$$

Proof. Since the pole orders at P_∞^n of any different b-reduced monomials are different, by the strong triangle inequality, the valuation at P_∞^n of a sum of b-reduced monomials is the minimum of the valuations of the monomials (i.e. there is no pole cancellation). □

Therefore, to calculate the possible pole orders of all polynomials in Γ, it is enough to

calculate the possible pole orders of monomials in Γ.

3.3.2 Valuations of rational functions

Before we can consider the possible pole orders of rational functions, we need a proposition from [Ful69].

Proposition 4. *For k algebraically closed, let I be an ideal in $k[x_1, \ldots, x_n]$. Suppose $V(I) = \{P_1, \ldots, P_m\}$ is a finite set of points in k^n. For $\mathcal{O}_i = \mathcal{O}_{P_i}$, there is a natural isomorphism*

$$k[x_1, \ldots, x_n]/I \longrightarrow \prod_{i=1}^{m} \mathcal{O}_i/I\mathcal{O}_i.$$

Theorem 5. *Let N be the semi-group of pole orders at P_∞^n in F_n generated by elements of Γ. Suppose, for some $r > 0$ and $r \notin N$, that there exists $\psi \in F_n$ with $(\psi)_\infty = rP_\infty^n$. Then, for $\psi = g/h$ with $g, h \in \Gamma$, there is a place in the support of $(h)_0$ corresponding to a singular point Q on C_n.*

Proof. 5 Fix some algebraic closure k of $\mathbb{F}_q$. Let $I_n = (f(x_0, x_1), \ldots, f(x_{n-1}, x_n)) \subset k[x_0, \ldots, x_n]$ be an ideal. Let $\Gamma = k[x_0, \ldots, x_n]/I_n$ be the polynomial ring of C_n.

Suppose there is an element $\psi \in F_n$ with pole only at Q_n. Then $\psi = g/h$ for polynomials $g, h \in \Gamma$. To prove the contrapositive, assume each zero P of h corresponds to a non-singular point of C_n. Let the local ring of P be $\mathcal{O}_P \subset F_n$ with valuation v_P. Since g/h does not have a pole at P,

$$v_P(g) \geq v_P(h) > 0.$$

Since P is non-singular, $\mathcal{O}_P$ is a valuation ring, so there is a parameter $t_P \in \mathcal{O}_P$ such that $v_P(t_P) = 1$. Since each element $z \in \mathcal{O}_P$ can be written uniquely as $z = t^k u$ for $u \in \mathcal{O}_P^\times$ and $k = v_P(z)$, and since $v_P(g) \geq v_P(h)$, we have

$$g \in (h) \subset \mathcal{O}_P.$$

Thus, for the ideal $I = (h, I_n)$, we have

$$g = 0 \in \mathcal{O}_P / I\mathcal{O}_P$$

for each zero P of h. Since there are finitely many zeroes of h in C_n, $V(I)$ contains a finite number of points. By the isomorphism from the above proposition,

$$g = 0 \in k[x_0, \ldots, x_n]/I = \Gamma/(h).$$

Hence, $g = \phi \cdot h \in \Gamma$ for some $\phi \in \Gamma$, so $\psi = g/h = \phi$, a polynomial. While ϕ may have coefficients in an extension of $\mathbb{F}_q$, the valuation of ϕ is that of a polynomial, and thus in N. $\square$

Thus, if C_n is non-singular, the set of pole orders of rational functions is the set of pole orders of monomials.

A non-example

We give an example of a singular C_{ab} curve with a rational function in the Riemann-Roch space.

Consider the curve $C \subset \overline{\mathbb{F}_q}^2$, for q odd, given by

$$C : y^2 - x^2(x-1) = 0.$$

In the associated function field $F/\mathbb{F}_q$, the functions x and y have poles of orders 2 and 3 at P_∞ and nowhere else. Let $P_{i,j}$ be the place corresponding to the point $(i, j) \in C$. Then the divisors associated to x and y are

$$\begin{aligned}(x) &= 2P_{0,0} - 2P_\infty \\ (y) &= 2P_{0,0} + P_{1,0} - 3P_\infty.\end{aligned}$$

Thus,

$$(y/x) = P_{1,0} - P_\infty,$$

so y/x has a pole that is not the pole of a monomial in x and y. There is one place in the support of $(x)_0$, which corresponds to the singular point $(0,0) \in C$, as is expected by the theorem. Note that $\mathcal{O}_{P_{0,0}}$ is not a valuation ring because there is no parameter. In particular, y is not in the ideal generated by x.

3.3.3 Gap numbers of monomials

We now calculate the number of Weierstrass gap numbers at P_∞^n by looking at the pole orders of monomials. We will assume that the curve C_n is non-singular. (If C_n is singular, we only obtain an upper bound for the genus by counting pole orders of monomials.)

For the function field F_n, the elements

$$1, x_0, x_1, \ldots, x_n$$

have poles at P_∞^n of orders

$$0, b^n, b^{n-1}a, \ldots, a^n,$$

respectively. The monomial $x_0^{e_0} x_1^{e_1} \ldots x_n^{e_n}$ has a pole of order $e_0 b^n + e_1 b^{n-1} a + \cdots + e_n a^n$. Thus, to calculate the genus, we need to calculate the number of positive integers α that

do not have a solution in non-negative integers e_i to the equation

$$\alpha = e_0 b^n + e_1 b^{n-1} a + \cdots + e_n a^n. \tag{3.1}$$

As we saw in Section 3.3.1, we can restrict to the case where $0 \leq e_i < b$.

Proposition 6. *Let $g(F_n)$ denote the genus of the function field F_n. For $n \geq 0$,*

$$g(F_{n+1}) = bg(F_n) + \frac{(a^{n+1} - 1)(b-1)}{2}.$$

Proof. To prove this, we will use two methods of counting, show that the methods do not overlap, and then show that we have not missed anything.

Method 1: Let $A = \{\alpha_i : i = 1, 2, \ldots, g(F_n)\}$ be the set of gap numbers of P_∞^n in F_n. We want to show that for any $\alpha \in A$, there is no monomial with a pole of order $b\alpha + la^{n+1}$ in F_{n+1} for $l \in R_b = \{0, \ldots, b-1\}$.

Since α is a gap number of P_∞^n in F_n, this means that there is no solution in e_i to equation (3.1).

Suppose we have a monomial with pole order equal to $b\alpha + la^{n+1}$ in F_{n+1}. The variables in F_{n+1} are $1, x_0, \ldots, x_{n+1}$, so this would mean we could find a solution to

$$e_0 b^{n+1} + e_1 ab^n + \cdots + e_n a^n b + e_{n+1} a^{n+1} = b\alpha + la^{n+1}$$

with non-negative integers e_i. Then $e_{n+1} \equiv l \mod b$. Since e_{n+1} and l are in R_b, $e_{n+1} = l$, so

$$b\alpha = e_0 b^{n+1} + e_1 ab^n + \cdots + e_n a^n b.$$

Dividing through by b, one has

$$\alpha = e_0 b^n + e_1 ab^{n-1} + \cdots + e_n a^n.$$

Since $e_0, e_1, \ldots, e_{n-1}$, and e_n are all non-negative integers, this contradicts the fact that α is a gap number of P_∞^n.

The result is that we have gap numbers $b\alpha + la^{n+1}$ of P_∞^{n+1} in F_{n+1} for $l = 0, \ldots, b-1$ and for all $\alpha \in A$. This gives us the $bg(F_n)$ term in the genus formula.

Method 2: Since the only pole orders below a^{n+1} that are achievable with monomials in F_{n+1} are multiples of b, there is no monomial with pole order that is congruent to any of $1, 2, \ldots, b-1 \mod b$ and less than a^{n+1}. Then, between a^{n+1} and $2a^{n+1}$, the only pole orders that are achievable are congruent to 0 or $a^{n+1} \mod b$. Going on in this manner, for $l = 1, 2, \ldots, b-1$, between la^{n+1} and $(l+1)a^{n+1}$, the only pole orders that are achievable are congruent to $0, a^{n+1}, 2a^{n+1}, \ldots, la^{n+1} \mod b$. (Note that these are all distinct congruence classes because a^{n+1} is a unit mod b.)

Counting in this way gives that the number of gap numbers from 0 to la^{n+1} that are congruent to $la^{n+1} \mod b$ is $\lfloor \frac{la^n}{b} \rfloor$. Let p_{n+1} be the total number of these gap numbers in F_{n+1}. For the $b-1$ non-zero congruent classes modulo b, we get

$$p_{n+1} = \left\lfloor \frac{a^{n+1}}{b} \right\rfloor + \left\lfloor \frac{2a^{n+1}}{b} \right\rfloor + \cdots + \left\lfloor \frac{(b-1)a^{n+1}}{b} \right\rfloor$$

missing pole orders.

For any integer m, let $\overline{m} \in R_b$ be the residue of m modulo b. Since

$$\left\lfloor \frac{m}{b} \right\rfloor = \frac{m}{b} - \frac{\overline{m}}{b},$$

and since $\left\{\overline{a^{n+1}}, \overline{2a^{n+1}}, \ldots, \overline{(b-1)a^{n+1}}\right\} = \{1, 2, \ldots, b-1\}$, we get that

$$\begin{aligned} p_{n+1} &= \left(\frac{a^{n+1} + 2a^{n+1} + \cdots + (b-1)a^{n+1}}{b} - \frac{1 + 2 + \cdots + (b-1)}{b} \right) \\ &= \frac{(a^{n+1} - 1)(b-1)}{2}. \end{aligned}$$

Overlap? Suppose there exists an element in F_{n+1} with pole order β that is counted

by both methods. By the first method, $\beta = b\alpha + la^{n+1}$ for some $\alpha \in A$ and l such that $0 \leq l \leq b-1$. So $\beta \equiv la^{n+1} \mod b$. If β is counted by the second method, then since $\beta \equiv la^{n+1}$, β must be less than la^{n+1}. But $\beta = b\alpha + la^{n+1}$, so this is a contradiction. So

$$g(F_{n+1}) \geq bg(F_n) + \frac{(a^{n+1}-1)(b-1)}{2}.$$

Everything? Suppose there exists a gap number γ for which there is no monomial in F_{n+1} with pole order equal to γ. Then $\gamma \equiv la^{n+1} \mod b$ for some $l \in \{0, 1, \dots, b-1\}$. Since $v_{\infty,n+1}(x_{n+1}^l) = -la^{n+1}$, we have $\gamma \neq la^{n+1}$. Thus, either $\gamma > la^{n+1}$ or $\gamma < la^{n+1}$.

If $\gamma > la^{n+1}$, then since $\gamma \equiv la^{n+1} \mod b$,

$$\gamma = mb + la^{n+1}$$

for some positive integer m. It follows that mb is a gap number of P_∞^{n+1}, because if it were the pole order of a function f, then the pole order of $f \cdot x_{n+1}^l$ is γ. Thus, m is a gap number of P_∞^n, and we counted all of these gap numbers in the first method.

Suppose $\gamma < la^{n+1}$. In the second method, we counted all integers that are congruent to la^{n+1} modulo b that are less than la^{n+1}, so γ must have been among them. $\square$

We have our result.

Proposition 7. *Let q be a power of a prime. For $f(x,y) \in \mathbb{F}_q[x,y]$, let $f = 0$ be the equation of a C_{ab} curve such that* $\gcd(a,b) = 1$. *For $n \geq 0$, let*

$$C_n = \left\{(p_0, \dots, p_n) \in \mathbb{F}_q^{n+1} : f(p_{i-1}, p_i) = 0 \text{ for } i = 1, \dots, n\right\}.$$

Consider the tower of function fields $\mathcal{F} = (F_0, F_1, \dots)$ where F_n is the function field associated to C_n. If C_n is non-singular, the genus of F_n is given by

$$g(F_n) = \frac{(b-1)a^{n+1} - (a-1)b^{n+1} + a - b}{2(a-b)}.$$

3.4 Practicality

We have seen that for the towers defined recursively by C_{ab} equations, the genus grows exponentially in a. The number of places of degree one will grow at most exponentially in b, which is strictly smaller than a. Thus, the ratio of number of places of degree one of F_n divided by the genus of F_n will tend to zero as $n \to \infty$. Therefore, these towers are not *asymptotically good*.

However, the non-singular curves in these towers have the property that the linear space of functions associated to the divisor kP_∞^n is readily calculable for every n: a basis of $\mathcal{L}(kP_\infty^n)$ is

$$\left\{ x_0^{e_0} \dots x_n^{e_n} : e_0 \geq 0, 0 \leq e_i < b \text{ for } i = 1, \dots, n \text{ and } \sum_{i=0}^{n} e_i a^i b^{n-i} \leq k \right\}.$$

Provided one can find towers with sufficiently many places of degree one in the first few levels, one can then create codes with good properties quite easily.

Chapter 4

Effective construction

In [GV02], van der Geer and van der Vlugt gave equations for an asymptotically good tower $\mathcal{C}$ of curves over $\mathbb{F}_8$ with $\lambda(\mathcal{C}) = 3/2$. In this chapter, we will briefly describe how they calculated $\lambda(\mathcal{C})$ and then see a way to construct the codes corresponding to each level of the tower.

4.1 The tower

The tower is described as follows: Let C be the closure of the affine curve in $\mathbb{P}^1 \times \mathbb{P}^1$ over $\mathbb{F}_2$ given by

$$y^2 + y = x + 1 + 1/x,$$

and let

$$D_i = \left\{(p_0, \dots, p_i) \in \mathbb{P}^1 \times \cdots \times \mathbb{P}^1 : (p_j, p_{j+1}) \in C \text{ for } j = 0, 1, \dots, i-1\right\}.$$

For C_i the normalization of D_i, we consider the tower $\mathcal{C} = (C_1, C_2, \dots)$.

Let $F_0 = \mathbb{F}_2(x_0)$. For $i > 0$, the function field F_i associated to D_i (and thus C_i) is

$$F_i = F_{i-1}(x_i),$$

where

$$x_i^2 + x_i = x_{i-1} + 1 + 1/x_{i-1}.$$

One can represent a point $P \in C_i$ by $(a_0, a_1, \ldots, a_i)$, although this representation is not necessarily unique for points with coordinates in $\mathbb{P}(\mathbb{F}_4)$.[1] For i an even integer, the zeroes of x_i occur at points $(a_0, a_1, \ldots, a_{i-1}, 0)$ where $a_{even} = 1$ and $a_{odd} \in \mathbb{F}_4 - \{0, 1\}$ (for i odd, we have $a_{even} \in \mathbb{F}_4 - \{0, 1\}$ and $a_{odd} = 1$). The poles of x_i occur at points $(a_0, a_1, \ldots, a_{i-j}, \infty^j)$ for $j > 0$ where $(a_0, a_1, \ldots, a_{i-j})$ is a zero of x_{i-j} and ∞^j denotes j consecutive coordinates of ∞.

In order to calculate the genus of C_i, the authors calculate the ramification divisor in each level of the tower. For each point $P \in C_i$ that is a pole or zero of x_i, they need to understand the order of vanishing of $f_i = x_i + 1 + 1/x_i$ at P. They do so by considering rational functions on C_i as a formal power series in a local parameter.

There is a natural projection $\pi_j : C_j \to C_{j-1}$ which maps $P_j = (a_0, \ldots, a_j) \in C_j$ to $P_{j-1} = (a_0, \ldots, a_{j-1}) \in C_{j-1}$. Consider a sequence of points $P_0 \in C_0$, $P_1 \in C_1$, $\ldots$, $P_i \in C_i$, where $\pi_j(P_j) = P_{j-1}$ for $j = 1, \ldots, i$. Let $t = x_0 + a_0$ be a local parameter at P_0 on C_0. The pullback of this is a local parameter at P_j on C_j. The function x_j can be written as a power series in t as

$$x_j = a_j + m_j(t),$$

where $m_j(t) \in \bar{\mathbb{F}}_2[[t]]$, a formal power series ring, and $m_j(0) = 0$.

Before we go on, it is necessary to know about Artin-Schreier extensions (taken from

[1] The singular points on C_i have coordinates in $\mathbb{P}(\mathbb{F}_4)$, as is seen in Appendix B, so multiple points in the normalization C_i can correspond to the same point in D_i.

[Sti93], Proposition III.7.8), and in fact, we will only use properties 2 and 3.

Proposition 8. *(Artin-Schreier Extensions) Let F/K be an algebraic function field of characteristic $p > 0$. For $u \in F$ such that $u \neq w^p - w$ for all $w \in F$, let $F' = F(y)$ with $y^p - y = u$. This is an Artin-Schreier extension.*

For $P \in \mathbb{P}_F$, define $m_P \in \mathbb{Z}$ by

$$m_P = \begin{cases} m, & \text{if } v_P(u - (z^p - z)) = -m < 0, p \nmid m, \text{ for some } z \in F. \\ -1, & \text{if } v_P(u - (z^p - z)) \geq 0 \text{ for some } z \in F. \end{cases}$$

Then:

1. *F'/F is a Galois extension of degree p (and thus cyclic), and the Galois elements are given by $\sigma(y) = y + v$ for $v = 0, 1, \ldots, p-1$.*

2. *P is unramified in F'/F iff $m_P = -1$.*

3. *P is totally ramified in F'/F iff $m_P > 0$. For P' the unique place over P, we have $d(P'|P) = (p-1)(m_P + 1)$.*

4. *If at least one place $Q \in \mathbb{P}_F$ has $m_Q > 0$, then K is algebraically closed in F (i.e. $K' = K$) and*

$$g' = p \cdot g + \frac{p-1}{2}\left(-2 + \sum_{P \in \mathbb{P}_F} (m_P + 1) \deg P\right)$$

While working in characteristic 2, for $z \in F_i$, let $\wp(z) = z^2 + z$ $(= z^p - z)$. Assuming i is even (the results are similar for i odd), one can then rewrite the equation for the extension C_{i+1}/C_i at $P_i = (1, a_1, \ldots, a_{i-1}, 0)$

$$x_{i+1}^2 + x_{i+1} = x_i + 1 + \frac{1}{x_i}$$

as

$$x_{i+1}^2 + x_{i+1} = 1/t + \wp(\gamma) + \mathcal{O}(P_{i+1})$$

where $\gamma \in \mathbf{F}(t)$ and $\mathcal{O}(P_{i+1})$ denotes a regular function $\beta(t)$ at P_{i+1}, i.e., $v_{P_{i+1}}(\beta(t)) \geq 0$.

Thus,

$$v_{P_{i+i}}(x_{i+i}^2 + x_{i+i} - \wp(\gamma(t))) = -1,$$

so this place is totally ramified in the extension C_{i+1}/C_i.

In the next step, they consider the pole $P_{i+1} = (1, a_1, \ldots, a_{i-1}, 0, \infty)$ in C_{i+2}/C_{i+1}. The equation

$$x_{i+2}^2 + x_{i+2} = x_{i+1} + 1 + \frac{1}{x_{i+1}}$$

is reduced to

$$x_{i+2}^2 + x_{i+2} = \wp(\gamma(t)) + \mathcal{O}(P_{i+1}),$$

for some $\gamma(t) \in \mathbf{F}(t)$. Thus,

$$v_{P_{i+2}}(x_{i+2}^2 + x_{i+2} - \wp(\gamma(t))) \geq 0,$$

so P_{i+1} is unramified in C_{i+2}/C_{i+1}.

These points are singular, so in some levels of the tower, they split into two points, which makes the extension unramified. However, since each point can only split into a finite number of points, eventually they finish splitting and ramify in every level. Thus, after a certain level (depending on i), they find the points $(a_0, a_1, \ldots, a_{i-1}, 0, \infty, \ldots, \infty)$ are ramified in every level of the tower.

x	y
0	∞
∞	∞
1	ρ, ρ^2
ρ, ρ^2	$0, 1$

Table 4.1: Behavior of points (x, y) on $C : y^2 + y = x + 1 + 1/x$ with coordinates in $\mathbb{P}(\mathbb{F}_4)$, with $\mathbb{F}_4 \cong \mathbb{F}_2(\rho)$, where $\rho^2 + \rho + 1 = 0$.

Working in this way, they are able to determine exactly which poles and zeroes are ramified or unramified, and what their contribution to the ramification divisor is. They can also calculate how many ramified points there are. Once they have this, they can apply the Hurwitz genus formula to calculate the genus for every level of the tower.

Finally, one sees that the first level of the curve has many $\mathbb{F}_8$-rational points (as described in the following section) that split completely in every level of the tower. The end result is that the tower is asymptotically good with

$$\lim_{i \to \infty} \frac{\#C_i(\mathbb{F}_8)}{g(C_i)} = \frac{3}{2}.$$

4.2 Creating codes

As was mentioned in the previous section, the only points in the tower that are ramified have coordinates in $\mathbb{F}_4$. In Table 4.1, the sequences of coordinate values for ramified points are given.

The curve C_1 has 14 $\mathbb{F}_8$-rational points. Twelve of these points have coordinates in $\mathbb{F}_8 - \{0, 1\}$, and they split completely in every level of the tower, giving us $6 \cdot 2^n$ $\mathbb{F}_8$-rational points in the nth level of the tower. In Table 4.2, the sequences of coordinate values for

x	y
α	α^4, α^6
α^2	α, α^5
α^3	α^2, α^3
α^4	α^2, α^3
α^5	α, α^5
α^6	α^4, α^6

Table 4.2: Behavior of points (x, y) on $C : y^2 + y = x + 1 + 1/x$ with coordinates in $\mathbb{F}_8 \cong \mathbb{F}_2(\alpha)$, where $\alpha^3 + \alpha^2 + 1 = 0$.

these points are given.

Throughout this paper, when working with this tower, we will let ρ and α be extension elements of $\mathbb{F}_2$ with $\rho^2 + \rho + 1 = 0$ and $\alpha^3 + \alpha^2 + 1 = 0$ (so $\mathbb{F}_4 \cong \mathbb{F}_2(\rho)$ and $\mathbb{F}_8 \cong \mathbb{F}_2(\alpha)$).

In order to create a code $\mathcal{C}_{\mathcal{L}}(C_i, \mathcal{P}, D)$, one needs to calculate a basis of the Riemann-Roch space associated to D and evaluate the basis elements at points in $\mathcal{P}$. For notation, in this section, let $P_{a_0,\ldots,a_i}$ denote the point(s) $(a_0, \ldots, a_i) \in C_i$, and let ∞^k denote k consecutive coordinates of ∞. There is a unique point $P_{\infty^{i+1}} \in C_i$, and we will take our divisor D to be $D = nP_{\infty^{i+1}}$. The set $\mathcal{P}$ will consist of the $6 \cdot 2^i$ points with coordinates in $\mathbb{F}_8 - \{0, 1\}$.

4.2.1 Level one

The first level of the curve is given by the equation

$$y^2 + y = x + 1 + 1/x.$$

We will consider divisors associated to functions on the curve. The support of $(x)_0$ consists of all points that have first coordinate equal to 0. This is just $P_{0,\infty}$. Similarly, the support

of $(x)_\infty$ contains only P_{∞^2}. Both of these places are ramified over $\mathbb{F}_2(x)$, so $(x)_0 = 2P_{0,\infty}$ and $(x)_\infty = 2P_{\infty^2}$. Looking at zeroes and poles of y, one has $(y)_0 = P_{\rho,0} + P_{\rho^2,0}$ and $(y)_\infty = P_{0,\infty} + P_{\infty^2}$.

With this information, we can construct a basis for $\mathcal{L}(rP_{\infty^2})$. The genus of this curve is 1, so by Riemann-Roch, we have

$$\dim \mathcal{L}(rP_{\infty,\infty}) = r$$

for $r > 0$. The functions 1, x, and xy have (respectively) pole orders of 0, 2, and 3. These functions will multiplicatively generate all pole orders greater than 1, so we have calculated a basis of $\mathcal{L}(rP_{\infty^2})$ for all $n > 1$.

It is then straightforward to evaluate these functions at the 12 points with coordinates in $\mathbb{F}_8 - \{0, 1\}$ to create the encoding matrix for our code.

4.2.2 Level two

The second level of the curve is given by two equations in variables x, y, and z:

$$y^2 + y = x + 1 + 1/x,$$

$$z^2 + z = y + 1 + 1/y.$$

Proceeding as before, we find

$$\begin{aligned}
(x) &= 4P_{0,\infty^2} - 4P_{\infty^3}, \\
(y) &= 2P_{\rho,0,\infty} + 2P_{\rho^2,0,\infty} - 2P_{0,\infty^2} - 2P_{\infty^3}, \\
(z) &= 2P_{1,\rho,0} + 2P_{1,\rho^2,0} - P_{\rho,0,\infty} - P_{\rho^2,0,\infty} - P_{0,\infty^2} - P_{\infty^3}.
\end{aligned}$$

The genus of this curve is 5, so, by Riemann-Roch,

$$\dim \mathcal{L}(9P_{\infty,\infty,\infty}) = 5.$$

Certainly, the functions $1, x, xy, xyz$, and x^2 are in this space, with pole orders (resp.) of 0, 4, 6, 7, and 8, and one can see that these functions will generate all pole orders above 10. As before, it is straightforward to evaluate these functions at the points with coordinates in $\mathbb{F}_8 - \{0, 1\}$, so one can create the encoding matrix associated to the code.

4.2.3 Level three

We now have four variables, x, y, z, and w with associated divisors

$$\begin{aligned}
(x) &= 8P_{0,\infty^3} - 8P_{\infty^4}, \\
(y) &= 4P_{\rho,0,\infty^2} + 4P_{\rho^2,0,\infty^2} - 4P_{0,\infty^3} - 4P_{\infty^4}, \\
(z) &= 4P_{1,\rho,0,\infty} + 4P_{1,\rho^2,0,\infty} - 2P_{\rho,0,\infty^2} - 2P_{\rho^2,0,\infty^2} \\
&\quad -2P_{0,\infty^3} - 2P_{\infty^4}, \\
(w) &= 2P_{\rho,1,\rho,0} + 2P_{\rho^2,1,\rho,0} + 2P_{\rho,1,\rho^2,0} + 2P_{\rho^2,1,\rho^2,0} - 2P_{1,\rho,0,\infty} \\
&\quad -2P_{1,\rho^2,0,\infty} - P_{\rho,0,\infty^2} - P_{\rho^2,0,\infty^2} - P_{0,\infty^3} - P_{\infty^4}.
\end{aligned}$$

The genus is 15, so we calculate a basis for $\mathcal{L}(29P_{\infty^4})$. We have the functions 1, x, xy, xyz, $xyzw$, x^2, x^2y, x^2yz, x^2yzw, x^3, x^2y^2z, x^2y^2zw, x^3y, and $x^2y^2z^2w$ with pole orders (resp.) of 0, 8, 12, 14, 15, 16, 20, 22, 23, 24, 26, 27, 28, and 29. The dimension of $\mathcal{L}(29P_{\infty^4})$ is 15, but we have only found 14 basis elements with monomials.

In order to calculate the fifteenth basis element, we look for functions that have zeroes at points where x, y, z, and w have poles (other than at P_{∞^4}). Some of these functions are $x+1$, x^2+x+1, and y^2+y+1. One finds

$$\begin{aligned}
(x+1) &= P_{1,\rho,1,\rho} + P_{1,\rho,1,\rho^2} + P_{1,\rho^2,1,\rho} + P_{1,\rho^2,1,\rho^2} \\
&\quad +2P_{1,\rho,0,\infty} + 2P_{1,\rho^2,0,\infty} - 8P_{\infty^4}.
\end{aligned}$$

The poles of the function xyw (aside from at P_{∞^4}) are cancelled by the zeroes of $x+1$. The

product $x(x+1)yw$ has a pole of order 21 at P_{∞^4}, and so is our fifteenth basis element.

4.2.4 Level four and beyond

As we can see from the previous section, a basis of monomials is not going to be sufficient to calculate a basis for $\mathcal{L}(rP_{\infty^{i+1}})$ for higher levels. In the third level, we needed to cancel unwanted poles, so we multiplied by low-degree polynomials (in one variable) that had zeroes of a high enough degree. This lead us to the function $x(x+1)yw$ which had a pole order that we had not previously found.

In the fourth level, the genus is 37, so the space $\mathcal{L}(73P_{\infty^5})$ has dimension 37. With variables x, y, z, w, and v, we find 27 monomials in the basis. As before, we look at the poles in all of our monomials and search for polynomials with zeroes to cancel these poles. The simplest of these functions are $x+1$, x^2+x+1, $y+1$, y^2+y+1, and z^2+z+1.

Using these simple polynomials to cancel unwanted poles from our monomials, we obtain 6 more functions for our basis. However, having looked at all possible monomials multiplied by all combinations of these polynomials, we have only found 33 out of the 37 functions.

What went wrong? We are working with the curves C_i which are the normalizations of the curves D_i. D_1 and D_2 are smooth, but D_i, for $i \geq 3$, is not. The presence of singular points (as described in Appendix B) causes the genus to drop, which means there are fewer Weierstrass gap numbers, so there are more functions in $\mathcal{L}(rP_{\infty^{i+1}})$.

Since our monomials have poles at $P_{\infty^{i+1}}$, there are only a finite number of monomials that have a pole of order $\leq r$ at $P_{\infty^{i+1}}$. Thus, one can find all products of monomials with

simple polynomials in $\mathcal{L}(rP_{\infty^{i+1}})$ in a finite amount of time. Since this does not produce a basis, the additional basis elements must be either less-easily factorizable polynomials or rational functions. The rational functions, in particular, are difficult to systematically find because there are pole cancellations.

In our experience, asymptotically good towers always have singularities. The question then arises: Given an asymptotically good tower, is there an algorithmic way to find a basis for $\mathcal{L}(rP)$ on each level of the tower?

4.3 Obtaining plane curves

Consider the function field F_n associated to the nth level of the tower. The goal of this section is to find a plane curve with associated function field F_n. Since there is a unique nonsingular curve associated to every function field of transcendence degree 1, the code created from the normalization of the plane curve will be equivalent to the code created from the normalization of the nth level of the tower. The motivation for writing the curve as a plane curve comes from [HB95] in which algorithms are given to create a code from a plane curve.

One way to find the plane curve is to find a primitive element for F_n over F_0, i.e., an element $\alpha_n \in F_n$ such that $F_n \cong F_0(\alpha_n)$. With α_n and its associated minimal polynomial $f_n(y) \in F_0[y] = \mathbb{F}_2(x_0)[y]$ over F_0, we have a plane curve in variables x_0 and y given by $f_n(y) = 0$. We can then clear the denominators of any x_0 terms to wind up with an element $\phi_n(x_0, y) \in \mathbb{F}_2[x_0, y]$. Once we have ϕ_n, we can use [HB95] to construct the associated Goppa code.

4.3.1 Primitive elements associated to the tower over $\mathbb{F}_8$

Claim: There exists $\phi_n(x,y) \in \mathbb{F}_2[x,y]$ such that $F_n \cong F_0(x_n)/(\phi_n(x_0,x_n))$. In other words, x_n is a primitive element for F_n/F_0.

Proof. By induction on n.

For $n = 1$,

$$\begin{aligned} F_1 &= \mathbb{F}_2(x_0,x_1)/(x_1^2 + x_1 + x_0 + 1 + \frac{1}{x_0}) \\ &= \mathbb{F}_2(x_0,x_1)/(x_0x_1^2 + x_0x_1 + x_0^2 + x_0 + 1). \end{aligned}$$

So $\phi_1(x,y) = xy^2 + xy + x^2 + x + 1 \in \mathbb{F}_2[x,y]$. The statement is true for $n = 1$.

Now, suppose the statement is true for $n = k-1$. Then there is a polynomial $\phi_{k-1}(x,y) \in \mathbb{F}_2[x,y]$ such that

$$F_{k-1} \cong \mathbb{F}_2(x_0,x_{k-1})/(\phi_{k-1}(x_0,x_{k-1})).$$

We have the field isomorphism

$$\begin{array}{rcccc} \pi & : & \mathbb{F}_2(x_0,\ldots,x_{k-1}) & \to & \mathbb{F}_2(x_1,\ldots,x_k) \\ & & x_i & \mapsto & x_{i+1}. \end{array}$$

Thus, since

$$\phi_{k-1}(x_0,x_{k-1}) = 0,$$

we also have

$$\phi_{k-1}(x_1,x_k) = 0.$$

Since

$$x_1^2 = x_1 + x_0 + 1 + \frac{1}{x_0}, \tag{4.1}$$

we can reduce all powers of x_1 to be elements of $\mathbb{F}_2(x_0) + x_1\mathbb{F}_2(x_0)$. Thus,

$$\phi_{k-1}(x_1, x_k) = f(x_0, x_k) + x_1 \cdot g(x_0, x_k),$$

for $f(x, y), g(x, y) \in \mathbb{F}_2(x, y)$. In particular,

$$x_1 = \frac{f(x_0, x_k)}{g(x_0, x_k)},$$

so $x_1 \in \mathbb{F}_2(x_0, x_k)$. Using (4.1) from above, we have

$$\begin{aligned} x_0 + 1 + \frac{1}{x_0} &= x_1^2 + x_1 \\ &= \left(\frac{f(x_0, x_k)}{g(x_0, x_k)}\right)^2 + \frac{f(x_0, x_k)}{g(x_0, x_k)}. \end{aligned}$$

Thus,

$$x_0 + 1 + \frac{1}{x_0} + \left(\frac{f(x_0, x_k)}{g(x_0, x_k)}\right)^2 + \frac{f(x_0, x_k)}{g(x_0, x_k)} = 0,$$

so we have a relation between x_0 and x_k. We clear denominators and let the resulting minimal polynomial be $\phi_k(x_0, x_k)$.

By induction,

$$\mathbb{F}_2(x_0, x_{k-1}) \cong F_{k-1} = \mathbb{F}_2(x_0, \ldots, x_{k-1}).$$

Since $\mathbb{F}_2(x_0, \ldots, x_{k-1})$ is isomorphic to $\mathbb{F}_2(x_1, \ldots, x_k)$, we have

$$\mathbb{F}_2(x_1, x_k) \cong \mathbb{F}_2(x_1, \ldots, x_k).$$

From above, $x_1 \in \mathbb{F}_2(x_0, x_k)$, so

$$\begin{aligned}
\mathbb{F}_2(x_0, x_k) &\cong \mathbb{F}_2(x_0, x_1, x_k) \\
&\cong \mathbb{F}_2(x_0)(x_1, x_k) \\
&\cong \mathbb{F}_2(x_0)(x_1, \ldots, x_k) \text{ (by induction)} \\
&\cong F_k.
\end{aligned}$$

Thus, we have found $\phi_k(x, y) \in \mathbb{F}_2[x, y]$ such that

$$F_k \cong \mathbb{F}_2(x_0, x_k)/(\phi_k(x_0, x_k)),$$

and so the claim is true for all positive integers n. $\square$

Equations of the associated plane curves

Using the method described above, one can obtain equations for plane curves associated to any level of the tower. The first few levels of the tower are given by:

$$\begin{aligned}
\phi_1(x, y) &= xy^2 + xy + x^2 + x + 1; \\
\phi_2(x, y) &= x^3y^4 + x^2y^4 + xy^4 + x^4 + x^3y + x^2y^2 + xy + 1; \\
\phi_3(x, y) &= x^7y^8 + x^6y^8 + x^5y^8 + x^6y^6 + x^7y^4 + x^6y^5 + x^5y^6 + x^3y^8 \\
&\quad + x^5y^5 + x^2y^8 + x^7y^2 + x^6y^3 + x^3y^6 + xy^8 + x^8 + x^7y \\
&\quad + x^5y^3 + x^4y^4 + x^3y^5 + x^2y^6 + x^2y^5 + x^4y^2 + x^3y^3 \\
&\quad + x^2y^3 + xy^4 + x^4 + xy^2 + xy + 1.
\end{aligned}$$

Note that the equations get large quickly, with $\deg(\phi_n) = 2^{n+1} - 1$.

4.4 Effective construction

In general, the difficulty in creating an algebraic geometry code lies in finding a basis for $\mathcal{L}(D)$ (as we saw in Section 4.2.3). In the case that the curve is a plane curve, however, there is an algorithm described by Haché and Le Brigand in [HB95] to create the associated algebraic geometry code. The algorithm uses blowing-ups to obtain a normalization of the curve. Once this is done, using Max Noether's "$H = AF + BG$" theorem, they give an algorithm in which the adjunction divisor is used to find a basis for $\mathcal{L}(D)$.

In this section, we will first see a brief overview of how the algorithm works. Afterwards, we will use the algorithm to create the code coming from the third level of the tower.

4.4.1 Preliminaries

Let $k = \mathbb{F}_q$, $\bar{k}$ an algebraic closure of k, C an affine curve defined and absolutely irreducible over k, and P a k-rational point on C. Under a linear change of coordinates, we can assume $P = (0, 0)$ and C is given by $f(x, y) = 0$. The coordinate ring of C is

$$\Gamma(C) = k[x, y]/(f),$$

and the function field of C is

$$k(C) = \{g/h : g, h \in \Gamma(C), h(P) \neq 0\},$$

that is, the quotient field of $\Gamma(C)$.

Now, let $\mathcal{C}$ be a projective plane curve defined by an absolutely irreducible equation

$F = 0$ for $F \in k[X, Y, Z]$, and let $\tilde{\mathcal{C}}$ be the smooth model of $\mathcal{C}$. The coordinate ring of $\mathcal{C}$ is

$$\Gamma(\mathcal{C}) = k[X, Y, Z]/(F).$$

For a form $G \in k[X, Y, Z]$, we let $\overline{G}$ denote its image in $\Gamma(\mathcal{C})$. Then the function field of $\mathcal{C}$ is

$$k(\mathcal{C}) = \{\overline{G}/\overline{H} : G, H \in k[X, Y, Z], \deg G = \deg H, \overline{H} \neq 0\},$$

and this is isomorphic to the function field K/k of $\tilde{\mathcal{C}}$.

For a point $P \in \mathcal{C}$, we change coordinates to $(X : Y : Z)$ so that $P = (0 : 0 : 1)$. For $f(x, y) = F(x, y, 1)$, the affine part $\mathcal{C}_P$ of $\mathcal{C}$ at P is given by the equation $f = 0$. $\mathcal{C}_P$ has associated function field $k(\mathcal{C}_P)$, which is naturally isomorphic to $k(\mathcal{C})$.

The local ring of $\mathcal{C}$ at P is

$$\mathcal{O}_P(\mathcal{C}) = \{g/h : g, h \in \Gamma(\mathcal{C}_P), h(P) \neq 0\},$$

and the unique maximal ideal is

$$\mathcal{M}_P(\mathcal{C}) = \{g/h \in \mathcal{O}_P(\mathcal{C}) : g(P) = 0\}.$$

For Q any k-rational point of $\mathcal{C}$ and $G \in k[X, Y, Z]$ a form of degree d, we map G into $\mathcal{O}_Q(\mathcal{C})$ as follows. Suppose Q is in the open set with $Z \neq 0$. Then we let

$$\overline{G}^Q = \overline{G}/\overline{Z}^d.$$

Singular points

Let F_X, F_Y, and F_Z denote the partial derivatives of F with respect to X, Y, and Z. Then a point $P \in \mathbb{P}^2(\bar{k})$ is singular if and only if

$$F(P) = F_X(P) = F_Y(P) = F_Z(P) = 0.$$

Such a system of equations can be solved using greatest common divisors and Gröbner bases.

For a point $P \in \mathcal{C}$, consider the affine curve $\mathcal{C}_P$ with equation $f(x, y) = 0$ and $P = (0, 0)$. For

$$f(x, y) = f_r(x, y) + f_{r+1}(x, y) + \cdots + f_n(x, y),$$

with f_i homogeneous forms of degree i, we define $m_P(f) = r$ to be the multiplicity of P on f. If $r = 1$, then P is nonsingular. If $r > 1$, P has a singularity of degree r (i.e., all partial derivatives of f up to the $(r-1)$st degree vanish at P).

Desingularization

In order to resolve the singularity of degree r (for $r \geq 1$) at a point $P \in \mathcal{C} \subset \mathbb{P}^2$, we use "blow-ups". We choose affine coordinates x and y so that $P = (0, 0)$ on the affine curve $C = \mathcal{C}_P$, given by the equation $f(x, y) = 0$.

Consider the subvariety of $C \times \mathbb{P}^1$ given by

$$\hat{C} = \big\{((x, y), (x_1 : y_1)) \in C \times \mathbb{P}^1 : xy_1 - yx_1 = 0\big\}$$

and the natural projection

$$\pi : \hat{C} \to C.$$

π is called the *blowing-up of P in C*.

For any point $(a, b) \in C$ with $a \neq 0$ and $b \neq 0$,

$$\pi^{-1}(a, b) = \{((a, b), (b : a))\}.$$

If $a \neq 0$ and $b = 0$,

$$\pi^{-1}(a, b) = \{((a, b), (0 : 1))\},$$

and if $a = 0$ and $b \neq 0$,

$$\pi^{-1}(a, b) = \{((a, b), (1 : 0))\}.$$

Finally,

$$\pi^{-1}(P) = \{(0, 0)\} \times \mathbb{P}^1.$$

We obtain an isomorphism between $C - \{P\}$ and $\hat{C} - \pi^{-1}(P)$.

Definitions 1. *For π the blowing-up morphism of P in C,*

- *The **total transform** of C by π is $\pi^{-1}(C)$.*
- *The **strict transform** C_1 of C by π is the closure of $\pi^{-1}(C - \{P\})$ in $\hat{C}$, that is, $C_1 = \overline{\pi^{-1}(C - \{P\})}$.*
- *The **exceptional line** $\mathcal{E}_1$ is $\pi^{-1}(P)$.*
- *The **infinitely near points of** P **by** π are the points in the intersection of C_1 and $\mathcal{E}_1$. If Q is an infinitely near point of P, we write $Q \succ P$.*

From the isomorphism above, C and C_1 are birationally equivalent. The singular points of C_1 are those of $C - \{P\}$ (using the identification of $C_1 - \pi^{-1}(P)$ with $C - \{P\}$) along with any infinitely near points of P. It is known that after a finite number of blowing-ups, one obtains a smooth curve C_n over k', for some finite extension k' of k, that is birationally equivalent to C. C_n is the smooth model of C.

The curve C_1 is covered by two open sets U_x and U_y, corresponding to $x_1 \neq 0$ and $y_1 \neq 0$. In U_x, we can take $x_1 = 1$. Points in this open set have $xy_1 = y$, so we can take local coordinates to be x and y_1. Then

$$f(x, y) = f(x, xy_1) = x^r f_x(x, y_1),$$

where r is the order of the singularity of P on C and $f_x(0, y_1)$ is not identically zero. The equation of the strict transform in U_x is $f_x(x, y_1) = 0$.

A point $Q = (x, y_1)$ is an infinitely near point of P if $x = 0$ and $f_x(x, y_1) = 0$, which is equivalent to

$$f_x(0, y_1) = 0.$$

Any root $y_1 = b_1 \in \bar{k}$ of the polynomial corresponds to an infinitely near point $Q = (0, b_1)$.[2] The local equation of Q in U_x is given by

$$f_x(x, y_1 - b_1) = 0.$$

We use a similar method to find infinitely near points in U_y. We take $y_1 = 1$, so

[2]The fact that $b_1 \in \bar{k}$ is the reason that the smooth model C_n of C may be defined over an extension of k.

$yx_1 = x$, so we have local coordinates y and x_1. We find

$$f(x, y) = f(x_1 y, y) = y^r f_y(x_1, y),$$

where r is the order of the singularity of P on $\mathcal{C}$ and $f_y(x_1, 0)$ is not identically zero. The infinitely near points correspond to roots of the polynomial

$$f_y(T, 0) \in \mathbb{F}_q[T].$$

However, since we obtain all infinitely near points with $x_1 = 1$ while working in U_x, we need only to check if $((x, y), (x_1 : y_1)) = ((0, 0), (0 : 1)) \in U_y$. Thus, we check to see if $f_y(0, 0) = 0$. If so, we have an additional infinitely near point $Q = ((0, 0), (0 : 1)) \in C_1$ with local equation given by $f_y(x_1, y) = 0$.

If any of the infinitely near points are singular, one can apply another blow-up. Proceeding with a finite number of blow-ups, one obtains a desingularization of the curve $\mathcal{C}$. For each singular point P on the curve, we obtain a "desingularization tree" $\mathcal{T}_P$. At P, we attach information about the local coordinates of P, the local equation, and the degree of the singularity. For every infinitely near point Q of P, there is a branch connecting P to Q, and $\mathcal{T}_P$ is constructed recursively - if any node Q is singular, then it is the root of a subtree $\mathcal{T}_Q$. When we get a nonsingular point, the branch stops, and this is a leaf. For each point Q in $\mathcal{T}_P$, we attach information about the local coordinates of Q, the exceptional line, the equation of the strict transform, and the degree of the singularity.

Monoidal transformations

With each blowup, we obtain the following isomorphisms:

$$\begin{array}{rccc} \pi_x^* : & k(C) & \to & k(C_1) \\ & g(x,y)/h(x,y) & \mapsto & g(x,xy_1)/h(x,xy_1) \end{array}$$

and

$$\begin{array}{rccc} \pi_y^* : & k(C) & \to & k(C_1) \\ & g(x,y)/h(x,y) & \mapsto & g(x_1y,y)/h(x_1y,y). \end{array}$$

π_x^* and π_y^* are called the *monoidal transformations* of the blowing-up of P. If $Q \succ P$ (for $Q \in U_x$, say), then

$$\pi_x^*(\mathcal{O}_P(C)) \subset \mathcal{O}_Q(C_1),$$

and similarly for π_y^* if $Q \in U_y$.

Local parametrizations

Let $\mathfrak{p}$ be a non-singular k_r-rational point on the smooth model of $\mathcal{C}$ for some extension k_r/k, where P is the unique k_r-rational point on $\mathcal{C}$ such that $\mathfrak{p} \succ P$ or $\mathfrak{p} = P$. Let $C_{\mathfrak{p}}$ be the affine plane curve at $\mathfrak{p} = (0,0)$ with equation $f(x,y) = 0$. Since $\mathfrak{p}$ is non-singular, there is a degree-one term in $f(x,y)$. Suppose there is a y-term. Then one can find a parametrization at $(0,0)$ in the formal power series ring $k_r[[t]]$ by setting

$$x(t) = t$$

and choosing appropriate $a_e \in k_r$ such that

$$y(t) = \sum_{e=1}^{\infty} a_e t^e.$$

In practice, only the first few terms of the expansion of $y(t)$ will be used. Note that if $\mathfrak{p}$ has a conjugate point $\mathfrak{p}'$ on $\mathcal{C}$, then the parametrization of $\mathcal{C}^{\mathfrak{p}'}$ can be obtained from the parametrization of $\mathcal{C}^{\mathfrak{p}}$ by conjugating a_e for $e = 1, 2, \ldots$.

Let $\phi^*_{\mathfrak{p}}$ denote the composition of the monoidal transformations and linear coordinate changes taking $k(C_P)$ to $k(C_{\mathfrak{p}})$. For any form $H \in k[X, Y, Z]$,

$$\overline{H}^P \in \Gamma(\mathcal{C}_P),$$

so

$$\phi^*_{\mathfrak{p}}(\overline{H}^P) \in \Gamma(\mathcal{C}_{\mathfrak{p}}).$$

Since P is uniquely determined by $\mathfrak{p}$, we will let

$$\Phi^*_{\mathfrak{p}}(H) = \phi^*_{\mathfrak{p}}(\overline{H}^P).$$

The parametrization of H at $\mathfrak{p}$ is

$$\Phi^*_{\mathfrak{p}}(H)(x(t), y(t)) = \sum_{e=0}^{\infty} b_e t^e,$$

with $b_e \in k_r$.

$(H)_{\mathfrak{p}}$, the order of vanishing of H at $\mathfrak{p}$, is then the smallest e such that $b_e \neq 0$. For

notation, we let

$$\Delta_{\mathfrak{p}}(H,n) = \sum_{e=0}^{n-1} b_e t^e.$$

Then,

$$(H)_{\mathfrak{p}} \geq n \iff \Delta_{\mathfrak{p}}(H,n) = 0.$$

Divisors associated to forms

In order to find the divisor associated to a form $G \in k[X,Y,Z]$, we first need to find all points in the intersection of the smooth model of $\mathcal{C}$ and $\{G=0\}$.[3] This gives us the points in the support of (G). For each point $\mathfrak{p}$, we can then find the local parametrizations of $\mathcal{C}_{\mathfrak{p}}$ and use them to find

$$\Phi_{\mathfrak{p}}^*(G)(x(t),y(t)),$$

which gives us the order of vanishing of G at $\mathfrak{p}$.

Adjunction divisor

Let $\pi_1,\ldots,\pi_n$ correspond to successive blowing-ups of k'-rational points $P_1, \ldots, P_n$ with singularities of degree $r_1,\ldots,r_n$ ($r_i > 1$) in the desingularization tree of $\mathcal{C}$. Associated to each blow-up is an exceptional line $\mathcal{E}_i = 0$ which corresponds to the open set in which we find infinitely-near points of P_i. Let E_i be the divisor of $k(\mathcal{C})$ associated to $\mathcal{E}_i$. E_i is called the *exceptional divisor of* π_i.

The points on the exceptional line $\mathcal{E}_i$ are exactly the leaves in the desingularization subtree of P_i, so these points make up the support of E_i.

[3]For our purposes, rather than working with arbitrary forms G, we will typically choose forms for which the support is mostly singular points.

The *adjunction divisor* is

$$\mathcal{A} = \sum_{i=1}^{n} (r_i - 1) E_i.$$

With the adjunction divisor, along with Max Noether's "H=AF+BG" theorem, Le Brigand and Risler have the following theorem (from [BR88]):[4]

Theorem 9. *Let $\mathcal{C} : \{F = 0\}$ be a projective plane curve defined and absolutely irreducible over k, $\mathcal{A}$ its adjunction divisor, and D and positive divisor. If G_0 is a form of degree d in $k[X, Y, Z]$ with $(G_0) \geq \mathcal{A} + D$, then $\mathcal{L}(D)$ is given by*

$$\left\{ \overline{G}/\overline{G}_0 : G \textit{ is a form of degree } d \textit{ in } k[X, Y, Z] \textit{ and } (G) \geq (G_0) - D \right\} \cup \{0\}.$$

In order to use this theorem, we need to be able to do the following:

- calculate the adjuncton divisor,
- find all forms of a certain degree with given divisor greater than a particular positive divisor.

We can calculate the adjunction divisor as follows. For a singular point $P \in \mathcal{C}$, a point Q_i in $\mathcal{T}_P$ coming from the blow-up of P_i via π_i, and leaves $\mathfrak{p}_1, \ldots, \mathfrak{p}_m$ in $\mathcal{T}_{Q_i}$, let $\pi^*_{Q_i}$ denote the monoidal transformation taking $k(\mathcal{C}_P)$ to $k(\mathcal{C}_{Q_i})$. Suppose the exceptional line of Q_i is given by $\hat{g} = 0$ for $\hat{g} \in k(\mathcal{C}_{Q_i})$. Since

$$\pi^*_{Q_i} : k(\mathcal{C}_P) \to k(\mathcal{C}_{Q_i})$$

is an isomorphism, $\hat{g}$ has a unique pre-image $g \in k(\mathcal{C}_P)$. Through a linear change of

[4]Since we are typically working with "one-point" codes, we only need this result. For the more general case where $D = D_+ - D_-$ for positive divisors D_+ and D_-, there is a similar result given in [HB95].

coordinates and homogenization, one can find a form $G \in k[X, Y, Z]$ such that $\overline{G}^P = g$. The divisor corresponding to the exceptional line of Q_i is then

$$E_i = (G)_{\mathfrak{p}_1}\mathfrak{p}_1 + \cdots + (G)_{\mathfrak{p}_m}\mathfrak{p}_m.$$

Interpolating forms

In order to find all forms of a degree d that have associated divisor greater than a certain divisor D, we solve a linear system, as follows. Suppose

$$D = \sum_{i=1}^{m} n_i P_i$$

for unique k'-rational points P_i of degree one. Let $\{H_l : l = 1, \ldots, s\}$ be set the set of all monomials of degree d (so $s = (d+1)(d+2)/2$). For a form $G \in k[X, Y, Z]$ of degree d, we have

$$G = \sum_{l=1}^{s} c_l H_l$$

for $c_l \in k$. In order that $(G) \geq D$, one needs to solve the system

$$\Delta_{P_i}(G, n_i) = 0 \text{ for all } i = 1, \ldots, m,$$

which is equivalent to finding a vector $(c_1, \ldots, c_s) \in k^s$ such that

$$\sum_{l=1}^{s} c_l \Delta_{P_i}(H_l, n_i) = 0 \text{ for all } i = 1, \ldots, m.$$

Evaluating functions

Let P be a non-singular point on the affine curve C with coordinates (x, y) such that $P = (0, 0)$, and let $u = g(x, y)/h(x, y)$ where g and h are the images of some G, $H \in k[x, y]$. There are three cases to consider:

- if $m_P(H) = 0$, then $u(P) = G(0, 0)/H(0, 0)$;
- if $m_P(G) = 0$ and $m_P(H) > 0$, then $u(P)$ is not defined;
- if $m_P(G) > 0$ and $m_P(H) > 0$, then we need to use a monoidal transformation.

In the third case, we blow up P into a unique infinitely-near point Q and apply our monoidal transformation to g and h:

$$\pi_Q^*(u) = \frac{x^{m_P(G)} g_Q}{x^{m_P(H)} h_Q}.$$

There are two cases to consider.

If $m_P(G) \geq m_P(H)$, then

$$\pi_Q^*(u) = \frac{x^{(m_P(G) - m_P(H))} g_Q}{h_Q},$$

and if $m_P(G) < m_P(H)$, then

$$\pi_Q^*(u) = \frac{g_Q}{x^{(m_P(H) - m_P(G))} h_Q}.$$

Since $u(P) = \pi_Q^*(u)(Q)$, we return to the three cases above to evaluate $\pi_Q^*(u)$ at Q. Eventually, after a finite number of transformations, we will end up in one of the first two cases.

4.4.2 Constructing the code for the third level of the tower over $\mathbb{F}_8$

In this section, we will demonstrate the algorithm with the plane curve we obtained in Section 4.3.1 corresponding to the third level of the tower. Let $\phi_3(X,Y,Z) \in \mathbb{F}_2[X,Y,Z]$ be given by

$$\begin{aligned}\phi_3(X,Y,Z) &= X^8Z^7 + X^7Y^8 + X^7Y^4Z^4 + X^7Y^2Z^6 + X^7YZ^7 + X^6Y^8Z \\ &\quad + X^6Y^6Z^3 + X^6Y^5Z^4 + X^6Y^3Z^6 + X^5Y^8Z^2 + X^5Y^6Z^4 \\ &\quad + X^5Y^5Z^5 + X^5Y^3Z^7 + X^4Y^4Z^7 + X^4Y^2Z^9 + X^4Z^{11} \\ &\quad + X^3Y^8Z^4 + X^3Y^6Z^6 + X^3Y^5Z^7 + X^3Y^3Z^9 + X^2Y^8Z^5 \\ &\quad + X^2Y^6Z^7 + X^2Y^5Z^8 + X^2Y^3Z^{10} + XY^8Z^6 + XY^4Z^{10} \\ &\quad + XY^2Z^{12} + XYZ^{13} + Z^{15}.\end{aligned}$$

We let $\mathcal{C}$ denote the absolutely irreducible projective plane curve given by $\phi_3(X,Y,Z) = 0$.

Our aim is to create a code using divisors on the desingularization of $\mathcal{C}$ that correspond to those in our work of Section 4.2.3. However, since $\mathcal{C}$ has singularities, we first need to calculate the normalization before we can choose our divisors D and $\mathcal{P}$.

Since we will need to work over $\mathbb{F}_4$ and $\mathbb{F}_8$, we again fix elements ρ and α such that $\rho^2 + \rho + 1 = 0$ and $\alpha^3 + \alpha^2 + 1 = 0$. All algorithms were implemented in Singular 2.0 ([GPS01]) with the Brill-Noether library ([ML01]).

Singular points

There are 20 singular points in the closure of $\mathbb{F}_2$ corresponding to the following points:

$P_1 = (1:0:0)$	$P_2 = (0:1:0)$	$P_3 = (\rho:1:1)$
$P_4 = (1:\rho:1)$	$P_5 = (\rho:0:1)$	$P_6 = (\alpha:\alpha^2:1)$
$P_7 = (\alpha:\alpha^3:1)$	$P_8 = (\alpha^3:\alpha:1)$	$P_9 = (\alpha^3:\alpha^5:1)$,

along with their conjugate points from the Galois action of $\mathbb{F}_4$ or $\mathbb{F}_8$ over $\mathbb{F}_2$ (denoted $P_3', P_4', P_5', P_6', P_6'', P_7', P_7'', P_8', P_8'', P_9'$, and P_9'').

Desingularization trees

We will demonstrate the blowing-up of the point $P_2 = (0:1:0)$. The affine equation of the curve, when Y=1, is $f_2(x,z) = 0$ where

$$\begin{aligned} f_2(x,z) &= x^8z^7 + x^4z^{11} + z^{15} + x^7z^7 + xz^{13} + x^7z^6 + x^4z^9 + xz^{12} \\ &\quad + x^6z^6 + x^5z^7 + x^3z^9 + x^2z^{10} + x^7z^4 + x^4z^7 + xz^{10} + x^6z^4 \\ &\quad + x^5z^5 + x^3z^7 + x^2z^8 + x^6z^3 + x^5z^4 + x^3z^6 + x^2z^7 + x^7 \\ &\quad + x^6z + x^5z^2 + x^3z^4 + x^2z^5 + xz^6, \end{aligned}$$

hence P_2 is singular of degree $r = 7$.

Working in U_x, we find that $f_2(x, xz_1) = x^7 f_{2,x}(x, z_1)$, where

$$\begin{aligned} f_{2,x}(0,T) &= T^6 + T^5 + T^4 + T^2 + T + 1 \\ &= (T^2 + T + 1)(T+1)^4. \end{aligned}$$

Thus, we have three infinitely-near points with coordinates $((x, z), (x_1 : z_1))$:

$$\begin{aligned} Q_{2,1} &= ((0,0),(1:\rho)), \\ Q_{2,2} &= ((0,0),(1:\rho^2)), \\ Q_{2,3} &= ((0,0),(1:1)). \end{aligned}$$

Working in U_z, we find that $f_2(x_1 z, z) = z^7 f_{2,z}(x_1, z)$, where $f_{2,z}(0,0) = 0$, so there is an additional infinitely-near point

$$Q_{2,4} = ((0,0),(0:1)).$$

The local equations of $Q_{2,1}$, $Q_{2,2}$, $Q_{2,3}$, and $Q_{2,4}$ are given by

$$f_{2,1}(x, z_1) = f_{2,x}(x, z_1 - \rho) = 0,$$

$$f_{2,2}(x, z_1) = f_{2,x}(x, z_1 - \rho^2) = 0,$$

$$f_{2,3} = f_{2,x}(x, z_1 - 1) = 0,$$

and

$$f_{2,4}(x_1, z) = f_{2,z}(x_1, z) = 0.$$

As can be seen in Table 4.3,[5] the functions $f_{2,1}(x, z_1)$, $f_{2,2}(x, z_1)$, and $f_{2,4}(x, z_1)$ each contain a linear term, so the points $Q_{2,1}, Q_{2,2}$, and $Q_{2,4}$ are nonsingular. However, $f_{2,3}(x, z_1)$ has a

[5]Note that $Q_{2,1}$ and $Q_{2,2}$ are conjugates over $\mathbb{F}_2$, so one obtains $f_{2,2}(x, z_1)$ by conjugating $f_{2,1}(x, z_1)$, so they have the same singularity behavior.

$$
\begin{aligned}
f_{2,1}(x,z_1) = \;& x^8z_1^{15} + \rho x^8z_1^{14} + \rho^2x^8z_1^{13} + x^8z_1^{12} + x^7z_1^{13} + \rho^2x^8z_1^{11} \\
& + \rho x^7z_1^{12} + x^8z_1^{10} + x^6z_1^{12} + \rho x^8z_1^9 + \rho^2x^8z_1^8 + \rho x^7z_1^9 \\
& + \rho x^8z_1^7 + \rho^2x^7z_1^8 + x^6z_1^9 + x^5z_1^{10} + \rho^2x^8z_1^6 + x^7z_1^7 + x^5z_1^9 \\
& + x^4z_1^{10} + x^8z_1^5 + \rho x^7z_1^6 + x^5z_1^8 + \rho x^8z_1^4 + x^6z_1^6 + x^5z_1^7 \\
& + \rho^2x^4z_1^8 + \rho^2x^5z_1^6 + x^4z_1^7 + x^3z_1^8 + \rho x^7z_1^3 + \rho^2x^5z_1^5 + \rho x^4z_1^6 \\
& + x^3z_1^7 + \rho^2x^7z_1^2 + \rho x^5z_1^4 + \rho^2x^4z_1^5 + \rho x^3z_1^6 + x^2z_1^7 + \rho x^6z_1^2 \\
& + \rho x^5z_1^3 + \rho x^3z_1^5 + \rho^2x^2z_1^6 + \rho^2x^6z_1 + \rho x^5z_1^2 + \rho x^4z_1^3 \\
& + \rho x^3z_1^4 + \rho^2x^2z_1^5 + x^6 + \rho x^5z_1 + \rho x^3z_1^3 + \rho^2x^2z_1^4 + z_1^6 \\
& + x^4z_1 + \rho^2x^3z_1^2 + \rho^2x^2z_1^3 + z_1^5 + \rho x^4 + \rho^2x^3z_1 + \rho^2x^2z_1^2 \\
& + \rho x^2z_1 + \rho^2z_1^2 + \rho^2z_1
\end{aligned}
$$

$$
\begin{aligned}
f_{2,3}(x,z_1) = \;& x^8z_1^{15} + x^8z_1^{14} + x^8z_1^{13} + x^8z_1^{12} + x^7z_1^{13} + x^7z_1^{12} + x^6z_1^{12} \\
& + x^7z_1^9 + x^7z_1^8 + x^6z_1^9 + x^5z_1^{10} + x^7z_1^7 + x^5z_1^9 + x^4z_1^{10} \\
& + x^7z_1^6 + x^6z_1^6 + x^5z_1^7 + x^4z_1^8 + x^8z_1^3 + x^4z_1^7 + x^3z_1^8 + x^8z_1^2 \\
& + x^7z_1^3 + x^5z_1^5 + x^4z_1^6 + x^3z_1^7 + x^8z_1 + x^7z_1^2 + x^4z_1^5 + x^3z_1^6 \\
& + x^2z_1^7 + x^8 + x^6z_1^2 + x^5z_1^3 + x^6z_1 + x^5z_1^2 + x^4z_1^3 + x^3z_1^4 \\
& + x^2z_1^5 + x^6 + x^3z_1^3 + x^2z_1^4 + z_1^6 + x^4z_1 + x^3z_1^2 + z_1^5 + x^4 \\
& + x^2z_1^2 + z_1^4
\end{aligned}
$$

$$
\begin{aligned}
f_{2,4}(x_1,z) = \;& x_1^8z^8 + x_1^7z^7 + x_1^7z^6 + x_1^4z^8 + x_1^7z^4 + x_1^6z^5 + x_1^5z^5 + x_1^4z^6 \\
& + x_1^6z^3 + x_1^6z^2 + x_1^5z^3 + x_1^4z^4 + x_1^3z^5 + x_1z^7 + z^8 + x_1^7 \\
& + x_1^5z^2 + x_1^2z^5 + x_1z^6 + x_1^6 + x_1^3z^3 + x_1^5 + x_1^3z^2 + x_1^2z^3 \\
& + x_1z^4 + x_1^2z^2 + x_1^3 + x_1^2 + x_1
\end{aligned}
$$

Table 4.3: Local equations of $Q_{2,1}$, $Q_{2,3}$, and $Q_{2,4}$

degree-4 singularity at $Q_{2,3}$, so we blow this point up. From another blow-up, $Q_{2,3}$ splits into two $\mathbb{F}_4$-rational (and conjugate) non-singular points $Q_{2,5}$ and $Q_{2,6}$.

The desingularization tree of P_2 can be found in Table 4.5. The desingularizations of the other singular points are given in Table 4.6 in less detail. When we are blowing up a point P_i, as above, we let $Q_{i,j}$ refer to the result of the jth blow-up.

$$
\begin{aligned}
f_{2,5}(x_2, z) = \quad & x^{19}z_2^{15} + \rho x^{19}z_2^{14} + \rho^2 x^{19}z_2^{13} + x^{18}z_2^{14} + x^{19}z_2^{12} + \rho x^{19}z_2^{11} \\
& + \rho^2 x^{18}z_2^{12} + x^{17}z_2^{13} + \rho^2 x^{19}z_2^{10} + \rho x^{17}z_2^{12} + x^{16}z_2^{13} + x^{19}z_2^{9} \\
& + \rho x^{18}z_2^{10} + \rho^2 x^{16}z_2^{12} + \rho x^{19}z_2^{8} + x^{15}z_2^{12} + \rho^2 x^{19}z_2^{7} + x^{18}z_2^{8} \\
& + \rho x^{17}z_2^{9} + x^{14}z_2^{12} + x^{19}z_2^{6} + \rho^2 x^{17}z_2^{8} + \rho x^{16}z_2^{9} + \rho x^{19}z_2^{5} \\
& + \rho^2 x^{18}z_2^{6} + x^{16}z_2^{8} + \rho^2 x^{19}z_2^{4} + \rho x^{15}z_2^{8} + x^{19}z_2^{3} + \rho x^{18}z_2^{4} \\
& + \rho^2 x^{17}z_2^{5} + \rho x^{14}z_2^{8} + \rho x^{19}z_2^{2} + x^{17}z_2^{4} + \rho^2 x^{16}z_2^{5} + x^{12}z_2^{9} \\
& + x^{11}z_2^{10} + \rho^2 x^{19}z_2 + x^{18}z_2^{2} + \rho x^{16}z_2^{4} + \rho x^{12}z_2^{8} + x^{11}z_2^{9} \\
& + x^{10}z_2^{10} + x^{19} + \rho^2 x^{15}z_2^{4} + x^{10}z_2^{9} + \rho^2 x^{18} + x^{17}z_2 \\
& + \rho^2 x^{14}z_2^{4} + x^{10}z_2^{8} + \rho x^{17} + x^{16}z_2 + x^{10}z_2^{7} + \rho^2 x^{16} + \rho x^{10}z_2^{6} \\
& + x^{8}z_2^{8} + x^{15} + \rho^2 x^{10}z_2^{5} + x^{9}z_2^{6} + x^{8}z_2^{7} + x^{7}z_2^{8} + x^{14} + x^{10}z_2^{4} \\
& + \rho^2 x^{8}z_2^{6} + x^{7}z_2^{7} + \rho^2 x^{12}z_2 + \rho^2 x^{11}z_2^{2} + \rho x^{10}z_2^{3} + \rho^2 x^{9}z_2^{4} \\
& + \rho^2 x^{8}z_2^{5} + \rho x^{7}z_2^{6} + x^{6}z_2^{7} + x^{12} + \rho^2 x^{11}z_2 + \rho x^{8}z_2^{4} + \rho^2 x^{7}z_2^{5} \\
& + \rho^2 x^{6}z_2^{6} + x^{5}z_2^{7} + \rho x^{10}z_2 + \rho x^{9}z_2^{2} + \rho x^{8}z_2^{3} + x^{7}z_2^{4} + \rho x^{6}z_2^{5} \\
& + \rho^2 x^{5}z_2^{6} + x^{10} + x^{8}z_2^{2} + \rho^2 x^{7}z_2^{3} + \rho x^{5}z_2^{5} + x^{9} + x^{8}z_2 + x^{7}z_2^{2} \\
& + \rho^2 x^{6}z_2^{3} + \rho x^{7}z_2 + \rho x^{6}z_2^{2} + \rho x^{5}z_2^{3} + x^{3}z_2^{5} + x^{2}z_2^{6} + x^{4}z_2^{3} \\
& + \rho^2 x^{3}z_2^{4} + \rho x^{6} + \rho x^{5}z_2 + \rho^2 x^{4}z_2^{2} + x^{3}z_2^{3} + \rho x^{2}z_2^{4} + xz_2^{5} + x^{5} \\
& + \rho^2 x^{4}z_2 + \rho^2 x^{3}z_2^{2} + x^{2}z_2^{3} + \rho xz_2^{4} + \rho^2 x^{4} + z_2^{4} + x^{3} + \rho^2 x^{2}z_2 \\
& + xz_2^{2} + \rho^2 x^{2} + \rho^2 xz_2 + z_2^{2} + \rho x
\end{aligned}
$$

Table 4.4: Local equations of $Q_{2,5}$

$$
\begin{array}{ccccc}
 & & \mathbf{Q_{2,1}} & & \\
 & & (x, z_1) = (0, \rho) & & \\
 & & f_{2,1}(x, z_1) = 0 & & \\
 & & r = 1 & & \\
 & & \mathcal{E}_{2,1} : \{x = 0\} & & \\
 & \nearrow & & & \\
 & \begin{array}{c} x = x \\ z = xz_1 \end{array} & & & \\
\mathbf{P_2} & & \mathbf{Q_{2,3}} & \begin{array}{c} x = x \\ z_2 = xz_1 \\ \longrightarrow \end{array} & \mathbf{Q_{2,5}} \\
(x, z) = (0, 0) & \begin{array}{c} x = x \\ z = xz_1 \\ \longrightarrow \end{array} & (x, z_1) = (0, 1) & & (x, z_2) = (0, \rho) \\
f_2(x, z) = 0 & & f_{2,3}(x, z_1) = 0 & & f_{2,5}(x, z_2) = 0 \\
r = 7 & & r = 4 & & r = 1 \\
 & & \mathcal{E}_{2,3} : \{x = 0\} & & \mathcal{E}_{2,5} : \{x = 0\} \\
 & \begin{array}{c} x = x_1 z \\ z = z \\ \searrow \end{array} & & & \\
 & & \mathbf{Q_{2,4}} & & \\
 & & (x_1, z) = (0, 0) & & \\
 & & f_{2,4}(x_1, z) = 0 & & \\
 & & r = 1 & & \\
 & & \mathcal{E}_{2,4} : \{z = 0\} & &
\end{array}
$$

Table 4.5: Desingularization of $P_2 = (0 : 1 : 0)$

$P_1 : (y,z) = (0,0)$, $r = 7$	$\longrightarrow$	$Q_{1,1} : (y, z_1) = (0,0)$
$P_2 : (x,z) = (0,0)$, $r = 7$	$\longrightarrow$	$\begin{cases} Q_{2,1} : (x, z_1) = (0, \rho) \\ Q_{2,2} : (x, z_1) = (0, \rho^2) \\ Q_{2,4} : (x_1, z) = (0,0) \\ Q_{2,5} : (x, z_2) = (0, \rho) \\ Q_{2,6} : (x, z_2) = (0, \rho^2) \end{cases}$
$P_3 : (x,y) = (\rho, 1)$, $r = 2$	$\longrightarrow$	$\begin{cases} Q_{3,2} : (x, y_2) = (\rho, 1) \\ Q_{3,3} : (x, y_2) = (\rho, \rho^2) \end{cases}$
$P_4 : (x,y) = (1, \rho)$, $r = 2$	$\longrightarrow$	$\begin{cases} Q_{4,4} : (x, y_4) = (1, \rho) \\ Q_{4,5} : (x, y_4) = (1, \rho^2) \end{cases}$
$P_5 : (x,y) = (\rho, 0)$, $r = 2$	$\longrightarrow$	$\begin{cases} Q_{5,2} : (x, y_2) = (\rho, 1) \\ Q_{5,3} : (x, y_2) = (\rho, \rho^2) \end{cases}$
$P_6 : (x,y) = (\alpha, \alpha^2)$, $r = 2$	$\longrightarrow$	$\begin{cases} Q_{6,1} : (x, y_1) = (\alpha, \alpha) \\ Q_{6,2} : (x, y_1) = (\alpha, \alpha^3) \end{cases}$
$P_7 : (x,y) = (\alpha, \alpha^3)$, $r = 2$	$\longrightarrow$	$\begin{cases} Q_{7,1} : (x, y_1) = (\alpha, \alpha) \\ Q_{7,2} : (x, y_1) = (\alpha, \alpha^3) \end{cases}$
$P_8 : (x,y) = (\alpha^3, \alpha)$, $r = 2$	$\longrightarrow$	$\begin{cases} Q_{8,1} : (x, y_1) = (\alpha^3, \alpha^5) \\ Q_{8,2} : (x, y_1) = (\alpha^3, \alpha^6) \end{cases}$
$P_9 : (x,y) = (\alpha^3, \alpha^5)$, $r = 2$	$\longrightarrow$	$\begin{cases} Q_{9,1} : (x, y_1) = (\alpha^3, \alpha^5) \\ Q_{9,2} : (x, y_1) = (\alpha^3, \alpha^6) \end{cases}$

Table 4.6: Blow-ups of all singular points

Parametrizations

We will find the first few terms of the local parametrization at $Q_{2,4}$. The local equation in x_1 and z is given by $f_{2,4}(x_1, z)$ in Table 4.3.

There is an x_1-term, so we set $z(t) = t$. If $x_1 = 0$, then

$$f_{2,4}(0, t) = t^8,$$

so the first term of $x_1(t)$ is t^8. (To cancel terms, we should have the first term equal to $-t^8$, but we are in characteristic 2.) Then we see that

$$f_{2,4}(t^8, t) = t^{12} + \ldots,$$

so the next term in $x_1(t)$ is t^{12}. Thus,

$$x_1(t) = t^8 + t^{12} + \ldots.$$

The projective coordinates of $Q_{2,4}$ are $(X : Y : Z) = (0 : 1 : 0)$, the affine coordinates are $(x, z) = (0, 0)$ with $X = x$, $Y = 1$, and $Z = z$, and the coordinates of the blow-up are $(x_1, z) = (0, 0)$ with $x = x_1 z$. Thus, we have

$$\begin{aligned}
\Phi^*_{Q_{2,4}}(X) &= x_1(t) \cdot z(t) = t^9 + t^{13} + \ldots, \\
\Phi^*_{Q_{2,4}}(Y) &= 1, \\
\Phi^*_{Q_{2,4}}(Z) &= z(t) = t.
\end{aligned}$$

In a similar manner, one can find the parametrizations of X, Y, and Z at all points. These values are given in Table 4.7. Note that if P and P' are conjugate points, then the

corresponding parametrizations are conjugate as well.

Adjunction divisor

We can explicitly calculate the component of the adjunction divisor corresponding to P_2 using the desingularization tree given in Table 4.5 and the local parametrizations given in Table 4.7.

For $Q_{2,1}$, the local equation of the curve is given in coordinates x and z_1, where $\overline{X}^{P_2} = x$ and $\overline{Z}^{P_2} = z = xz_1$. The exceptional line is given by $\{x = 0\}$. Since $x = \overline{X}^{P_2}$, we find

$$\Phi^*_{Q_{2,1}}(X) = t$$

to see that the order of vanishing is 1, and hence the exceptional divisor $E_{2,1}$ is $1Q_{2,1}$. There is a degree-7 singularity at P_1, so the component of the adjunction divisor coming from $Q_{2,1}$ is $(7-1)Q_{2,1} = 6Q_{2,1}$. Since $Q_{2,2}$ is a conjugate, we also have $6Q_{2,2}$ in the adjunction divisor.

For $Q_{2,4}$, the exceptional divisor is given by $z = 0$, where $z = \overline{Z}^{P_2}$.

$$\Phi^*_{Q_{2,4}}(Z) = t,$$

so $E_{2,4} = 1Q_{2,4}$. Again, as there is a degree-7 singularity at P_2, we obtain $6Q_{2,4}$ in the adjunction divisor.

To calculate the contribution from $Q_{2,5}$, we first need to look at the exceptional line of

$\Phi^*_{Q_{1,1}}(X) = 1$ $\Phi^*_{Q_{1,1}}(Y) = t^7 + t^{11} + \dots$ $\Phi^*_{Q_{1,1}}(Z) = t^8 + t^{12} + \dots.$	$\Phi^*_{Q_{6,1}}(X) = \alpha + t$ $\Phi^*_{Q_{6,1}}(Y) = \alpha^2 + \alpha t + \dots$ $\Phi^*_{Q_{6,1}}(Z) = 1.$
$\Phi^*_{Q_{2,1}}(X) = t$ $\Phi^*_{Q_{2,1}}(Y) = 1$ $\Phi^*_{Q_{2,1}}(Z) = \rho t + \rho^2 t^5 + \dots.$	$\Phi^*_{Q_{6,2}}(X) = \alpha + t$ $\Phi^*_{Q_{6,2}}(Y) = \alpha^2 + \alpha^3 t + \dots$ $\Phi^*_{Q_{6,2}}(Z) = 1.$
$\Phi^*_{Q_{2,4}}(X) = t^9 + t^{13} + \dots$ $\Phi^*_{Q_{2,4}}(Y) = 1$ $\Phi^*_{Q_{2,4}}(Z) = t.$	$\Phi^*_{Q_{7,1}}(X) = \alpha + t$ $\Phi^*_{Q_{7,1}}(Y) = \alpha^3 + \alpha t + \dots$ $\Phi^*_{Q_{7,1}}(Z) = 1.$
$\Phi^*_{Q_{2,5}}(X) = \rho^2 t^2 + t^3$ $\Phi^*_{Q_{2,5}}(Y) = 1$ $\Phi^*_{Q_{2,5}}(Z) = \rho^2 t^2 + t^3 + \rho^2 t^4 + \dots$	$\Phi^*_{Q_{7,2}}(X) = \alpha + t$ $\Phi^*_{Q_{7,2}}(Y) = \alpha^3 + \alpha^3 t + \dots$ $\Phi^*_{Q_{7,2}}(Z) = 1.$
$\Phi^*_{Q_{3,2}}(X) = \rho + t$ $\Phi^*_{Q_{3,2}}(Y) = 1 + t^2 + \rho^2 t^3 + \dots$ $\Phi^*_{Q_{3,2}}(Z) = 1.$	$\Phi^*_{Q_{8,1}}(X) = \alpha^3 + t$ $\Phi^*_{Q_{8,1}}(Y) = \alpha + \alpha^5 t + \dots$ $\Phi^*_{Q_{8,1}}(Z) = 1.$
$\Phi^*_{Q_{3,3}}(X) = \rho + t$ $\Phi^*_{Q_{3,3}}(Y) = 1 + \rho^2 t^2 + \rho t^3 + \dots$ $\Phi^*_{Q_{3,3}}(Z) = 1.$	$\Phi^*_{Q_{8,2}}(X) = \alpha^3 + t$ $\Phi^*_{Q_{8,2}}(Y) = \alpha + \alpha^6 t + \dots$ $\Phi^*_{Q_{8,2}}(Z) = 1.$
$\Phi^*_{Q_{4,4}}(X) = 1 + t$ $\Phi^*_{Q_{4,4}}(Y) = \rho + \rho t^4 + \rho^2 t^6 + \dots$ $\Phi^*_{Q_{4,4}}(Z) = 1.$	$\Phi^*_{Q_{9,1}}(X) = \alpha^3 + t$ $\Phi^*_{Q_{9,1}}(Y) = \alpha^5 + \alpha^5 t + \dots$ $\Phi^*_{Q_{9,1}}(Z) = 1.$
$\Phi^*_{Q_{5,2}}(X) = \rho + t$ $\Phi^*_{Q_{5,2}}(Y) = t^2 + \rho^2 t^3 + \dots$ $\Phi^*_{Q_{5,2}}(Z) = 1.$	$\Phi^*_{Q_{9,2}}(X) = \alpha^3 + t$ $\Phi^*_{Q_{9,2}}(Y) = \alpha^5 + \alpha^6 t + \dots$ $\Phi^*_{Q_{9,2}}(Z) = 1.$
$\Phi^*_{Q_{5,3}}(X) = \rho + t$ $\Phi^*_{Q_{5,3}}(Y) = \rho^2 t^2 + \rho t^3 + \dots$ $\Phi^*_{Q_{5,3}}(Z) = 1.$	

Table 4.7: Local parametrizations. (Note that the parametrizations at $Q_{2,2}$, $Q_{2,6}$, and $Q_{4,5}$ are conjugates of those at $Q_{2,1}$, $Q_{2,5}$, and $Q_{4,4}$, respectively.)

$Q_{2,3}$, which is given by $\{x = 0\}$. Since $\overline{X}^{P_2} = x$ and

$$\Phi^*_{Q_{2,5}}(X) = \rho^2 t^2 + \ldots,$$

$E_{2,3} = 2Q_{2,5}$. The degree-7 singularity gives $12Q_{2,5}$ coming from $Q_{2,3}$. Then, in the local equation of $Q_{2,5}$, the exceptional line is again $\{x = 0\}$, so $E_{2,5} = 2Q_{2,5}$, and since there is a degree-4 singularity at $Q_{2,3}$, we obtain $6Q_{2,5}$. Thus, the component of $\mathcal{A}$ coming from $Q_{2,5}$ is $18Q_{2,5}$, and as $Q_{2,6}$ is a conjugate of $Q_{2,5}$, we have $18Q_{2,6}$ also.

The components of the adjunction divisor coming from the other singular points can be calculated similarly. To simplify notation, let the $\mathbb{F}_2$-rational divisors $R_1, \ldots, R_{11}$ be as follows:

$$\begin{aligned}
R_1 &= Q_{1,1}, \\
R_2 &= Q_{2,1} + Q_{2,2}, \\
R_3 &= Q_{2,4}, \\
R_4 &= Q_{2,5} + Q_{2,6}, \\
R_5 &= Q_{3,2} + Q_{3,3} + Q'_{3,2} + Q'_{3,3}, \\
R_6 &= Q_{4,4} + Q_{4,5} + Q'_{4,4} + Q'_{4,5}, \\
R_7 &= Q_{5,2} + Q_{5,3} + Q'_{5,2} + Q'_{5,3}, \\
R_8 &= Q_{6,1} + Q_{6,2} + Q'_{6,1} + Q'_{6,2} + Q''_{6,1} + Q''_{6,2}, \\
R_9 &= Q_{7,1} + Q_{7,2} + Q'_{7,1} + Q'_{7,2} + Q''_{7,1} + Q''_{7,2}, \\
R_{10} &= Q_{8,1} + Q_{8,2} + Q'_{8,1} + Q'_{8,2} + Q''_{8,1} + Q''_{8,2}, \\
R_{11} &= Q_{9,1} + Q_{9,2} + Q'_{9,1} + Q'_{9,2} + Q''_{9,1} + Q''_{9,2}.
\end{aligned}$$

The result is the degree-152 divisor

$$\begin{aligned}\mathcal{A} &= 42R_1 + 6R_2 + 6R_3 + 18R_4 + 2R_5 + 4R_6 + 2R_7 \\ &\quad + R_8 + R_9 + R_{10} + R_{11}.\end{aligned}$$

Choice of D and $\mathcal{P}$

Comparing the nonsingular model of $\mathcal{C}$ to the third level of the tower (as in Section 4.2.3), we have the following points corresponding to each other:[6]

Point on the plane curve model	Point on the tower
$Q_{1,1}$	$P_{\infty,\infty,\infty,\infty}$
$Q_{2,1}$ and $Q_{2,2}$	$P_{\rho,0,\infty,\infty}$ and $P_{\rho^2,0,\infty,\infty}$
$Q_{2,4}$	$P_{0,\infty,\infty,\infty}$
$Q_{2,5}$ and $Q_{2,6}$	$P_{1,\rho,0,\infty}$ and $P_{1,\rho^2,0,\infty}$

We will therefore take $D = nR_1$ $(= nQ_{1,1})$. However, rather than dealing with an arbitrary value for n, we will calculate a basis for $\mathcal{L}(29R_1)$ and try to use the functions in that basis to (multiplicatively) generate all functions with higher pole orders.

As for $\mathcal{P}$, we will take the points to be all of the points with affine coordinates in $\mathbb{F}_8 - \{0, 1\}$. There are 36 points of the form $(\alpha^i : \alpha^j : 1)$ on the curve. Twelve of these are singular points which split into the non-singular points in the support of $R_8+R_9+R_{10}+R_{11}$, and the other 24 are non-singular points. Let the divisor R_{12} denote the sum of these 24 non-singular points. Thus, our divisor $\mathcal{P} = R_8 + R_9 + R_{10} + R_{11} + R_{12}$ consists of 48 points

[6] All other points on the plane curve model have $Z = 1$, so we can use the X and Y coordinates of a point on the plane model to determine the x and w coordinates of the corresponding point in the tower. For instance, when $Z = 1$, the four points in R_5 have $(X, Y) = (\rho, 1)$ or $(\rho^2, 1)$, and these correspond to the four points $P_{\rho,1,\rho,1}$, $P_{\rho^2,1,\rho,1}$, $P_{\rho,1,\rho^2,1}$, and $P_{\rho^2,1,\rho^2,1}$.

with coordinates in $\mathbb{F}_8$ (corresponding to the points with coordinates in $\mathbb{F}_8 - \{0,1\}$ in the third level of the tower).

Divisors associated to forms

With the parametrizations of X, Y, and Z at all of the points in the adjunction divisor $\mathcal{A}$ and the divisor D, we can begin to search for a form G_0 with

$$\begin{aligned}(G_0) &\geq \mathcal{A} + D \\ &= 71R_1 + 6R_2 + 6R_3 + 18R_4 + 2R_5 + 4R_6 + 2R_7 \\ &\quad + R_8 + R_9 + R_{10} + R_{11}.\end{aligned}$$

For G_0 a form of degree d, (G_0) has degree $15d$. Since our code $C_{\mathcal{L}}(\mathcal{P}, D)$ will have $\deg(D) = 29$, and since $deg(\mathcal{A}) = 152$, we need a form G_0 with associated divisor degree at least 181. Thus, the degree of G_0 must be at least 13.

Using the parametrizations of all of the points in the support of $\mathcal{A} + D$, we can find the divisors associated to many forms (particularly those that vanish at singular points). Some of these divisors are given in Table 4.8. (Note that we have not completely calculated the divisors associated to $Y^3 + Y^2Z + Z^3$ and $Y^3 + YZ^2 + Z^3$ because the support of these divisors is not limited to the singular points of the curve.)

We find that for

$$G_0 = Y(Y+Z)(X+Z)^4 Z^2 (Y^3 + Y^2Z + Y^3)(Y^3 + YZ^2 + Y^3),$$

a degree-14 form, then

$$\begin{array}{rcl}
(Y) & = & 7R_1 + 2R_7 \\
(Z) & = & 8R_1 + R_2 + R_3 + 2R_4 \\
(X+Z) & = & R_2 + R_3 + 4R_4 + R_6 \\
(Y+Z) & = & 7R_1 + 2R_5 \\
(X^2 + XZ + Z^2) & = & 6R_2 + 2R_3 + 4R_4 + R_5 + R_7 \\
(Y^3 + Y^2Z + Z^3) & \geq & 21R_1 + R_8 + R_{10} \\
(Y^3 + YZ^2 + Z^3) & \geq & 21R_1 + R_9 + R_{11}
\end{array}$$

Table 4.8: Divisors associated to certain forms

$$\begin{aligned}
(G_0) \quad &\geq \quad 72R_1 + 6R_2 + 6R_3 + 20R_4 + 2R_5 + 4R_6 + 2R_7 \\
&\qquad + R_8 + R_9 + R_{10} + R_{11} \\
&\geq \quad \mathcal{A} + D.
\end{aligned}$$

In order to use Theorem 9 to find a basis for $\mathcal{L}(D)$, we need to know (G_0) exactly. Therefore, we need to completely calculate the divisors associated to the irreducible forms of degree 3 in G_0.

Let

$$H = (Y^3 + Y^2Z + Z^3)(Y^3 + YZ^2 + Z^3).$$

All of the points in the divisor R_{12} are in the support of (H). Combining this with the information in Table 4.8, we find

$$(H) \geq 42R_1 + R_8 + R_9 + R_{10} + R_{11} + R_{12}.$$

The right-hand side is a divisor of degree 90. Since H is a form of degree 6, (H) is a divisor of degree 90, so we have equality. Thus,

$$\begin{aligned}(G_0) &= 72R_1 + 6R_2 + 6R_3 + 20R_4 + 2R_5 + 4R_6 + 2R_7 \\ &\quad + R_8 + R_9 + R_{10} + R_{11} + R_{12}.\end{aligned}$$

Interpolating forms

Now that we have a form G_0 of degree 14 with $(G_0) \geq \mathcal{A} + D$, we can look for forms G of degree 14 with

$$\begin{aligned}(G) \geq (G_0) - D &= 43R_1 + 6R_2 + 6R_3 + 20R_4 + 2R_5 + 4R_6 + 2R_7 \\ &\quad + R_8 + R_9 + R_{10} + R_{11} + R_{12}.\end{aligned}$$

Let $\{H_l : l = 1, 2, \ldots, 120\}$ be the set of monomials of degree 14 in X, Y, and Z. For notation, since there are 66 unique points in the support of $(G_0) - D$, let

$$(G_0) - D = \sum_{i=1}^{66} n_i Q_i$$

with unique places Q_i and $n_i > 0$. Using our parametrizations from Section 4.4.2, we can calculate

$$\Delta_{Q_i}(H_l, n_i) = \sum_{e=0}^{n_i - 1} b_{i,l,e} t^e$$

for each i and l. Then, for each i, we create an $n_i \times 120$ matrix

$$M_i = (a_{j,k}),$$

where $a_{j,k} = b_{i,j,k-1}$.

A vector $v = (a_1, \ldots, a_{120})$ is in the nullspace of M_i if and only if $\Delta_i(G, n_i) = 0$ for $G = \sum_{l=1}^{120} a_l H_l$. However, there may not be a $\mathbb{F}_2$-rational basis for the nullspace.

We concatenate the matrices $M_1, M_2, \ldots, M_{66}$ vertically to form a matrix M. The nullspace of the matrix M is the intersection of the nullspaces of the matrices $M_1, \ldots, M_{66}$. Although M has entries that lie in $\mathbb{F}_4$ or $\mathbb{F}_8$, when we row reduce M to form M_{red}, we find that all entries in M_{red} lie in $\mathbb{F}_2$. The nullspace of M_{red} (which is the nullspace of M) is 15-dimensional, and the basis elements lie in $\mathbb{F}_2^{120}$.

For each vector $v_i = (a_{1,i}, \ldots, a_{120,i})$ in the nullspace of M, we have an associated degree-14 form

$$G_i = a_{1,i}H_1 + \cdots + a_{120,i}H_{120} \in \mathbb{F}_2[X, Y, Z]$$

with $(G_i) \geq (G_0) - D$. Thus,

$$\mathcal{L}(29Q_{1,1}) = \langle G_i/G_0 : i = 1, \ldots, 15 \rangle.$$

From the parametrizations of X, Y, and Z at $Q_{1,1}$, we can find the pole order of the function G_i/G_0 at $Q_{1,1}$. If two functions have the same pole order, we can subtract one from the other to obtain a new function with strictly smaller pole order, eventually winding up with 15 distinct pole orders. If we let the set I consist of the possible pole orders of elements in $\mathcal{L}(29Q_{1,1})$, then

$$I = \{0, 8, 12, 14, 15, 16, 20, 21, 22, 23, 24, 26, 27, 28, 29\}.$$

For each $i \in I$, let f_i be a basis element of $\mathcal{L}(29Q_{1,1})$ with pole order i. These functions can be found in Table 4.9.

Looking back at Section 4.2.3, we see that, with monomials, we were missing a function

Order	Function
0	1
8	x
12	$\frac{y}{g_0}\left(\ (1+y^3)(x^2+x^3+x^6)+(y^3+y^7)(x+x^4+x^5)\ \right)$
14	$\frac{1}{g_0}\left(\begin{array}{c}(y^3+y^4+y^5+y^6)(1+x^2+x^4+x^6)\\+x^3(1+y+y^2)+(x+x^5)(1+y^2+y^8)\end{array}\right)$
15	$\frac{y}{g_0}\left(\begin{array}{c}(y^3+y^4+y^5+y^6)(1+x^2+x^4+x^6)\\+(x+x^5)(y+y^2+y^7)+x^3(1+y+y^2)\end{array}\right)$
16	x^2
20	$\frac{y}{g_0}\left(\ (y^3+y^7+x^7(1+y^3)+x^4(1+y^7)\ \right)$
21	$\frac{1}{g_0}\left(\ (1+x^7)y^2(1+y^3)\ \right)$
22	$\frac{x}{g_0}\left(\begin{array}{c}(y^3+y^5+y^6+y^8)(1+x^2+x^6+x^4)\\+(1+y^2+y^4)(x+x^3+x^5)+x^4(1+y^8)\end{array}\right)$
23	$\frac{x}{g_0}\left(\ (1+y^3)(x+x^3+x^5)+(y^3+y^7)(1+x^2+x^4+x^6)\ \right)$
24	$\frac{1}{g_0}\left(\begin{array}{c}(y^3+y^4+y^5+y^6)(x^2+x^3+x^5+x^6)+1+x^8\\+x^7(y+y^2)+x^4(1+y+y^2)+x(y^2+y^8)+x^5(y+y^8)\end{array}\right)$
26	$\frac{1}{g_0}\left(\begin{array}{c}1+(y^3+y^4+y^5+y^5)(x^3+x^5)+x+x^7\\+(1+y^2+y^8)(1+x^2+x^4+x^6)+x^8y(1+y)\end{array}\right)$
27	$\frac{1}{g_0}\left(\begin{array}{c}(1+y^2+y^3+y^7+y^8)(x^2+x^3+x^5+x^6)+x(1+y^7)\\+(x^3+x^5)(y+y^8)+x^4(1+y^4+y^5+y^6)+y^6\\+x^8y(1+y+y^2)+x^7y^4(1+y+y^3)\end{array}\right)$
28	$\frac{xy}{g_0}\left(\ y^3+y^7+x^7(1+y^3)+x^4(1+y^7)\ \right)$
29	$\frac{1}{g_0}\left(\begin{array}{c}1+(y+y^3+y^4)(x^3+x^4+x^5)+x^4+y^7(x^3+x^5+x^7)\\+(y^2+y^5+y^6)(x^2+x^6+x^7+x^8)+x(y^6+y^7)\\+x^8(1+y^6)\end{array}\right)$

Table 4.9: Pole orders of basis elements for $\mathcal{L}(29Q_{1,1})$ with $x = X/Z$, $y = Y/Z$, and $g_0(x,y) = G_0(x,y,1) = (x+1)^4 y(y^7+1)$.

with a pole of order 21. In coordinates (x, y, z, w), we see that the function

$$\frac{(x^7+1)w(w^3+1)}{(x+1)^4(w^7+1)}$$

is in the Riemann-Roch space with a pole of order 21.[7]

Evaluating functions

Now that we have calculated a basis for $\mathcal{L}(D)$, we need to evaluate our functions at all points in the support of $\mathcal{P}$. Since all points in $\mathcal{P}$ have $Z = 1$, we will work in this open set with affine coordinates $x = X$ and $y = Y$.

We will evaluate the function

$$f_{21}(x, y) = (x^7+1)y(y^3+1)/(x^4+1)(y^7+1)$$

at the (nonsingular) point (α, α). For notation, let $g(x, y)$ and $h(x, y)$ denote the numerator and denominator (respectively) of $f_{21}(x, y)$.

Let $P = (\alpha, \alpha)$. One finds that $g(P) = 0$ and $h(P) = 0$, so to determine $f_{21}(P)$, we need to apply a monoidal transformation. Changing coordinates so that $P = (0, 0)$, we blow up P on the curve via the transform

$$x \to x; \qquad y \to xy_1.$$

There is a unique infinitely near point Q with $y_1 = 1$.

[7] In fact, using the relations between x, y, z, and w in the tower, one finds that this function is exactly the function $x(x+1)yw$ that we found in Section 4.2.3. Note that we have not cancelled the $(x+1)$ or $(w+1)$ factors in the numerator and denominator to make the function easier to present here.

We apply our monoidal transformation to $g(x, y)$. Let

$$g_P(x, y) = g(x - \alpha, y - \alpha).$$

Then

$$g_P(x, xy_1) = x^1 g_{P,x}(x, y_1).$$

We evaluate the function $g_{P,x}(x, y_1)$ at our point $Q = (0, 1)$ to get

$$g_{P,x}(0, 1) = \alpha^2.$$

Applying the monoidal transformation to $h(x, y)$, one similarly finds $h_{P,x}(x, y_1)$ and

$$h_{P,x}(0, 1) = \alpha^5.$$

Thus,

$$f_{21}(P) = \frac{g(P)}{h(P)} = \frac{g_{P,x}(Q)}{h_{P,x}(Q)} = \frac{\alpha^2}{\alpha^5} = \alpha^4.$$

The same method can be used for all functions $f \in \mathcal{L}(D)$ and all points P in the support of $\mathcal{P}$. If $f(P) = \frac{0}{0}$, we use a monoidal transformation. Otherwise, we have the value $f(P) \in \mathbb{F}_8$.

The results of evaluating the basis elements of $\mathcal{L}(29Q_{1,1})$ at the points lying over singular points can be found in Table 4.10, and at the nonsingular points in Table 4.11. These columns, along with the additional 16 columns (each) due to conjugation, combine to form the generator matrix for our code. The parameters of this code are

$$\begin{aligned} \text{length} &= 48 \\ \text{dimension} &= 15 \\ \text{distance} &\geq 19. \end{aligned}$$

In order to create a code with dimension $k > 15$, one can use the functions in $\mathcal{L}(29Q_{1,1})$ to (multiplicatively) generate functions with larger pole orders. (Note that we really can do this.) Given the function

$$f_l = f_{j_1} f_{j_2} \dots f_{j_n}$$

with $j_m \in I$ for $m = 1, 2, \dots, n$ (and $\sum j_m = l$), one can calculate $f_l(P)$ by

$$f_l(P) = f_{j_1}(P) f_{j_2}(P) \dots f_{j_n}(P).$$

Equivalently, to calculate the row in the encoding matrix corresponding to f_l, one can multiply coordinate-wise the rows corresponding to f_{j_1}, f_{j_2}, $\dots$, f_{j_n}. This will produce a code with the following parameters:

$$\begin{aligned} \text{length} &= 48 \\ \text{dimension} &= k \\ \text{distance} &\geq 34 - k. \end{aligned}$$

	$Q_{6,1}$	$Q_{6,2}$	$Q_{7,1}$	$Q_{7,2}$	$Q_{8,1}$	$Q_{8,2}$	$Q_{9,1}$	$Q_{9,2}$
f_0	1	1	1	1	1	1	1	1
f_8	α	α	α	α	α^3	α^3	α^3	α^3
f_{12}	α	0	α	0	α	α^4	α	α^4
f_{14}	α^3	0	α^3	0	α^4	0	α^4	0
f_{15}	α^2	α^4	α^5	α^3	1	α	α^5	α^4
f_{16}	α^2	α^2	α^2	α^2	α^6	α^6	α^6	α^6
f_{20}	α	α^2	α	α^2	α^5	0	α^5	0
f_{21}	1	α^5	α	α^6	α^2	α	α^6	α^5
f_{22}	α^5	0	α^5	0	α^6	1	α^6	1
f_{23}	α	α^5	1	α^4	α^2	α^4	α^5	1
f_{24}	α^5	α^4	α^5	α^4	α^6	0	α^6	0
f_{26}	0	α	0	α	α	α^6	α	α^6
f_{27}	α^2	α^6	0	α^3	α^4	α	α^2	1
f_{28}	α^2	α^3	α^2	α^3	α	0	α	0
f_{29}	0	α	α^3	1	α^4	1	α^4	α

Table 4.10: Basis elements of $\mathcal{L}(29Q_{1,1})$ evaluated at certain points in $\mathcal{P}$

	(α,α)	(α,α^4)	(α,α^5)	(α,α^6)	(α^3,α^2)	(α^3,α^3)	(α^3,α^4)	(α^3,α^6)
f_0	1	1	1	1	1	1	1	1
f_8	α	α	α	α	α^3	α^3	α^3	α^3
f_{12}	0	α	0	α	α	α	α^4	α^4
f_{14}	α^5	α^2	α^5	α^2	1	1	α^5	α^5
f_{15}	0	α^4	1	1	α	α^6	0	α^6
f_{16}	α^2	α^2	α^2	α^2	α^6	α^6	α^6	α^6
f_{20}	α^2	α	α^2	α	α^5	α^5	0	0
f_{21}	α^4	α^2	α	α^4	α^3	α^4	α^4	α^6
f_{22}	α^6	1	α^6	1	α	α	α^5	α^5
f_{23}	α^6	α^6	α^2	α^4	α	1	α	α^6
f_{24}	α^4	α^5	α^4	α^5	α^6	α^6	0	0
f_{26}	1	α^5	1	α^5	α^3	α^3	1	1
f_{27}	α^5	1	α^3	α^2	α^5	α^5	0	α^6
f_{28}	α^3	α^2	α^3	α^2	α	α	0	0
f_{29}	α^2	1	1	α^6	0	α^2	0	1

Table 4.11: Basis elements of $\mathcal{L}(29Q_{1,1})$ evaluated at certain points in $\mathcal{P}$

Appendix A

Function Fields

We present some facts about algebraic function fields (without proofs). An excellent reference is [Sti93].

For any field K, an *algebraic function field* F/K of transcendence degree 1 is a finite algebraic extension of $K(x)$ for some element $x \in F$ which is transcendental over K. A *valuation ring* of F/K is a ring $\mathcal{O} \subseteq F$ such that

1. $K \subsetneq \mathcal{O} \subsetneq F$, and

2. for any $z \in F$, if $z \notin \mathcal{O}$, then $z^{-1} \in \mathcal{O}$.

Each valuation ring $\mathcal{O}$ in F/K is a local ring with unique maximal ideal $P = \mathcal{O} \setminus \mathcal{O}^\times$, where $\mathcal{O}^\times$ denotes the unit group of $\mathcal{O}$. For each valuation (and thus local) ring $\mathcal{O}$, the maximal ideal P is called a *place* of F/K. Each place P is a principal ideal. Any $t \in \mathcal{O}$ for which $P = t\mathcal{O}$ is called a local parameter for P. Fixing a local parameter t for P, any element $z \in F^\times$ can be written uniquely as $z = t^n \cdot u$ for $u \in \mathcal{O}^\times$. This gives rise to a discrete valuation $v_P : F \to \mathbb{Z} \cup \{\infty\}$ where $v_P(z) = n$ and $v_P(0) = \infty$. $\mathcal{O}$ is then uniquely determined by P:

$$\mathcal{O} = \mathcal{O}_P = \{z \in F : v_P(z) \geq 0\}.$$

Let $\mathbb{P}_F$ denote the set of places in F. There are infinitely many places in any function field. For instance, when $F = K(x)$, a rational function field, there is a one-to-one correspondence between places of F/K and monic irreducible polynomials $p(x) \in K[x]$ along with the element $1/x$.

For $P \in \mathbb{P}_F$, $F_P = \mathcal{O}_P/P$ is the *residue class field* of P with the map from F to $F_P \cup \{\infty\}$, given by $z \mapsto z(P)$, called the *residue class map*. The *degree* of a place P is $\deg P = [F_P : K]$.

For any place P of degree one, $F_P = K$. We can regard any element $z \in F$ as a function that maps places of degree one to $K \cup \{\infty\}$ by $P \mapsto z(P)$. It is for this reason that F/K is called a *function field*.

For $P \in \mathbb{P}_F$ (of any degree) and $z \in F$, if $z(P) = 0$ (resp. $z(P) = \infty$), then P is a *zero* (resp. *pole*) of z.

A *divisor* D is a formal sum of places, given by

$$D = \sum_{P \in \mathbb{P}_F} n_P P,$$

for $n_P \in \mathbb{Z}$, with $n_P = 0$ for almost all P. The degree of D is given by

$$\deg D = \sum_{P \in \mathbb{P}_F} n_P (\deg P).$$

If $n_P \geq 0$ for all P, D is said to be *effective*, and this is denoted by $D \geq 0$. The divisor group $\mathcal{D}_F$ of F/K is the free abelian group generated by all places P of F/K.

For any $z \in F^\times$, let Z (resp. N) denote the set of zeroes (resp. poles) of z. Then

$$(z)_0 = \sum_{P \in Z} v_P(z)P, \text{ the zero-divisor of } z,$$

and

$$(z)_\infty = \sum_{P \in N} (-v_P(z))P, \text{ the pole-divisor of } z.$$

The divisor associated to z, called the *principal divisor* of z, is $(z) = (z)_0 - (z)_\infty$, ie,

$$(z) = \sum_{P \in \mathbb{P}_F} v_P(z)P.$$

The degree of any principal divisor is zero.

For any divisor $D \in \mathcal{D}_F$, we define the Riemann-Roch space

$$\mathcal{L}(D) = \{z \in F : (z) + D \geq 0\} \cup \{0\},$$

which is a vector space over K. In particular, if $\deg D < 0$, then $\mathcal{L}(D) = \{0\}$, and $\mathcal{L}(0) = K$.

There is a constant $\gamma \in \mathbb{Z}$ (depending only on F/K) such that

$$\deg D - \dim \mathcal{L}(D) \leq \gamma$$

for all $D \in \mathcal{D}_F$. With this in mind, one defines the *genus* g of F/K to be

$$g = \max\{\deg D - \dim \mathcal{L}(D) + 1 : D \in \mathcal{D}_F\}.$$

Note that $g \geq 0$ because for $D = 0$, one has $\deg 0 - \dim \mathcal{L}(0) + 1 = 0 - 1 + 1 = 0$.

Theorem 10. *(Riemann-Roch Theorem, I.5.15 in [Sti93]) Let W be a canonical divisor of F/K. For any $D \in \mathcal{D}_F$,*

$$\dim \mathcal{L}(D) = \deg D + 1 - g + \dim \mathcal{L}(W - D).$$

A canonical divisor W of F/K is the divisor associated to a Weil differential. All we need here is that for any canonical divisor W, one has $\deg W = 2g - 2$. Thus, if $\deg D > 2g - 2$, we have $\dim \mathcal{L}(W - D) = 0$, so

$$\dim \mathcal{L}(D) = \deg D + 1 - g.$$

If $P \in \mathbb{P}_F$ is a place of degree one, we consider the vector spaces

$$\mathcal{L}(0) \subseteq \mathcal{L}(P) \subseteq \mathcal{L}(2P) \subseteq \cdots \subseteq \mathcal{L}(nP) \subseteq \ldots.$$

For each $n > 0$, one can show that either

$$\dim \mathcal{L}(nP) = \dim \mathcal{L}((n-1)P),$$

or

$$\dim \mathcal{L}(nP) = \dim \mathcal{L}((n-1)P) + 1.$$

If $\dim \mathcal{L}(nP) = \dim \mathcal{L}((n-1)P)$, then $\mathcal{L}(nP) = \mathcal{L}((n-1)P)$, so there is no element $z \in F$ such that $(z)_\infty = nP$. In this case, we say n is a *Weierstrass gap number of* P. Otherwise, if $\dim \mathcal{L}(nP) = \dim \mathcal{L}((n-1)P) + 1$, there is an element $z \in F$ with $(z)_\infty = nP$, and we

say n is a *non-gap* of P.

The non-gaps form an additive semi-group. By the Riemann-Roch Theorem, for any place P of degree one, there are g gap numbers $n_1, \ldots, n_g$ with

$$1 = n_1 < n_2 < \cdots < n_g < 2g.$$

Appendix B

Identifying singular points

B.1 Determining singular points on D_i

Let $\mathcal{C} = (C_1, C_2, \dots)$ be the tower of curves defined over $\mathbb{F}_2$ given by van der Geer and van der Vlugt in Section 4.1. For C the closure of the affine curve in $\mathbb{P}^1 \times \mathbb{P}^1$ over $\mathbb{F}_2$ given by

$$y^2 + y = x + 1 + 1/x,$$

the curve D_i is given by

$$D_i = \left\{ (p_0, \dots, p_i) \in \mathbb{P}^1 \times \cdots \times \mathbb{P}^1 : (p_j, p_j + 1) \in C \text{ for } j = 0, \dots, i-1 \right\}.$$

Points on D_i satisfy the i equations

$$\begin{gathered} x_1^2 + x_1 = x_0 + 1 + 1/x_0, \\ \vdots \\ x_i^2 + x_i = x_{i-1} + 1 + 1/x_{i-1}. \end{gathered}$$

We can multiply through by denominators in each equation to obtain i simultaneous polynomial equations:

$$x_0x_1^2 + x_0x_1 + x_0^2 + x_0 + 1 = 0,$$

$$\vdots$$

$$x_{i-1}x_i^2 + x_{i-1}x_i + x_{i-1}^2 + x_{i-1} + 1 = 0.$$

Fixing an algebraic closure $\bar{\mathbb{F}}_2$, we can calculate the singular points with coordinates in $\bar{\mathbb{F}}_2 \times \cdots \times \bar{\mathbb{F}}_2$ with the Jacobian by calculating the partial derivatives of each equation:

$$J(x_0, \ldots, x_i) = \begin{pmatrix} x_1^2 + x_1 + 1 & x_0 & 0 & \ldots & 0 & 0 \\ 0 & x_2^2 + x_2 + 1 & x_1 & \ldots & 0 & 0 \\ \vdots & & & \ddots & & \\ 0 & 0 & 0 & \ldots & x_i^2 + x_i + 1 & x_{i-1} \end{pmatrix}.$$

A point $P \in C$ with $P = (p_0, \ldots, p_i) \in \bar{\mathbb{F}}_2 \times \cdots \times \bar{\mathbb{F}}_2$ is non-singular if $J(P)$ has maximal rank $(= i - 1)$. In order for the rank to drop, a point P will have to have coordinates p_j (for $j > 0$) and p_k (for $k < i$) such that $p_j^2 + p_j + 1 = 0$ and $p_k = 0$. As we see from Table 4.1, such points have $p_{j-1} = 1$ and $p_{k+1} = \infty$.

We now consider points that do not have all coordinates in $\bar{\mathbb{F}}_2$. For P, let k be the smallest integer such that $p_{k+1} = \infty$. By Table 4.1, all subsequent coordinates are ∞, and the preceding coordinates are in $\mathbb{F}_4$:

$$P = (p_0, \ldots, p_k, \infty, \infty, \ldots, \infty)$$

with $p_j \in \mathbb{F}_4$ for $j = 0, \ldots, k$.

To calculate the Jacobian of such a point P, we invert $x_{k+1}, \ldots, x_i$ in the defining polynomials and clear denominators:

$$x_0x_1^2 + x_0x_1 + x_0^2 + x_0 + 1 = 0,$$

$$\vdots$$

$$x_{k-1}x_k^2 + x_{k-1}x_k + x_{k-1}^2 + x_{k-1} + 1 = 0,$$

$$x_k + x_kx_k + x_k^2x_{k+1}^2 + x_kx_{k+1}^2 + x_{k+1}^2 = 0,$$

$$x_{k+1} + x_{k+1}x_{k+2} + x_{k+1}^2x_{k+2}^2 + x_{k+1}x_{k+2}^2 + x_{k+2}^2 = 0,$$

$$\vdots$$

$$x_{i-1} + x_{i-1}x_i + x_{i-1}^2x_i^2 + x_{i-1}x_i^2 + x_i^2.$$

The resulting Jacobian turns out to be the same as the Jacobian obtained before, except now when we evaluate at $J(P)$, we have $x_j = 0$ (corresponding to $p_j = \infty$) for $j \geq k$. The rank of the Jacobian drops when we have some point with $p_j^2 + p_j + 1 = 0$ for $j > 0$.

To summarize, the singular points in D_i are exactly those points P with some coordinate $p_j = 1$ and $p_i = \infty$. They are, for $a_j \in \mathbb{F}_4 - \{0, 1\}$, of the form

$$(a_0, 1, a_2, 1, \ldots, a_{2k}, 0, \infty, \ldots, \infty) \text{ for } 1 \leq k \leq \frac{i}{2} - 1$$

or

$$(1, a_1, 1, a_3, \ldots, 1, a_{2k-1}, 0, \infty, \ldots, \infty) \text{ for } 1 \leq k \leq \frac{i-1}{2}.$$

There are no singular points on D_1 or D_2. On D_3, the points $(1, \rho, 0, \infty)$ and $(1, \rho^2, 0, \infty)$ are singular, and so on.

Bibliography

[ADKS01] I. Aleshnikov, V. Deolalikar, V. Kumar, and H. Stichtenoth, *Towards a basis for the space of regular functions in a tower of function fields meeting the Drinfeld-Vladuţ bound*, Finite Fields and Applications (D. Jungnickel and H. Neiderreiter, eds.), Springer, Berlin-Heidelberg, 2001.

[BGS] J. Bezerra, A. Garcia, and H. Stichtenoth, *An explicit tower of function fields over cubic finite fields and Zink's lower bound*, Submitted to American Journal of Mathematics in April, 2004.

[BR88] D. Le Brigand and J. J. Risler, *Algorithme de Brill-Noether et codes de Goppa*, Bulletin de la Société Mathématique de France **116** (1988), 231–253.

[Deo00] V. Deolalikar, *Towers of function fields with extremal properties*, ArXiv Mathematics e-prints (2000).

[DV83] V. G. Drinfeld and S. G. Vlăduţ, *Number of points of an algebraic curve*, Functional Analysis and its Applications **17** (January-March, 1983), no. 1, 53–54.

[Elk97] N. Elkies, *Explicit modular towers*, Proceedings of the Thirty-Fifth Annual Allerton Conference on Communication, Control and Computing (T. Basar and A. Vardy, eds.), University of Illinois at Urbana-Champaign, 1997, pp. 23–32.

[Elk01] ———, *Explicit towers of Drinfeld modular curves*, Progress in Mathematics (2001), no. 202, 189–198.

[Ful69] W. Fulton, *Algebraic curves*, Benjamin, New York, 1969.

[GPS01] G.-M. Greuel, G. Pfister, and H. Schönemann, SINGULAR *2.0*, A Computer Algebra System for Polynomial Computations, Centre for Computer Algebra, University of Kaiserslautern, 2001, `http://www.singular.uni-kl.de`.

[GS95] A. Garcia and H. Stichtenoth, *A tower of Artin-Schreier extensions of function fields attaining the Drinfeld-Vladuţ bound*, Inventiones Mathematicae **121** (1995), 211–222.

[GS96a] ———, *Asymptotically good towers of function fields over finite fields*, Comptes rendus de l'Académie des sciences de Paris, Série I **322** (1996), 1067–1070.

[GS96b] ———, *On the asymptotic behaviour of some towers of function fields over finite fields*, Journal of Number Theory **61** (1996), no. 147, 248–273.

[GST97] A. Garcia, H. Stichtenoth, and M. Thomas, *On towers and composita of towers of function fields over finite fields*, Finite Fields and their Applications **3** (1997), 257–274.

[GV02] G. van der Geer and M. van der Vlugt, *An asymptotically good tower of curves over the field with eight elements*, Bulletins of the London Mathematical Society **34** (2002), no. 3, 291–300.

[HB95] G. Haché and D. Le Brigand, *Effective construction of algebraic geometry codes*, IEEE Transactions on Information Theory **41** (November, 1995), no. 6, 1615–1628.

[Mat98] R. Matsumoto, *The c_{ab} curve*, Available at http://rmatsumoto.org/cab.pdf, 1998.

[ML01] J. I. F. Martin and C. Lossen, `brnoeth.lib`, A Singular 2.0 library for applications to Algebraic Geometry codes, Centre for Computer Algebra, University of Kaiserslautern, 2001, `http://www.singular.uni-kl.de`.

[Mor91] C. Moreno, *Algebraic curves over finite fields*, Cambridge University Press, Cambridge/New York/Port Chester/Melbourne/Sydney, 1991.

[Ser83] J.-P. Serre, *Sur le nombre des points rationnels d'une courbe algébrique sur un corps fini.*, Comptes rendus de l'Académie des sciences de Paris, Série I **296** (1983), 397–402.

[Sti93] H. Stichtenoth, *Algebraic function fields and codes*, Springer-Verlag, Berlin/Heidelberg/New York, 1993.

[Sti95] ———, *Algebraic geometric codes*, Different Aspects of Coding Theory (R. Calderbank, ed.), Proceedings of Symposia in Applied Mathematics, vol. 50, American Mathematical Society, Providence, Rhode Island, 1995, pp. 139–152.

[VH97] C. Voss and T. Høholdt, *An explicit construction of a sequence of codes attaining the Tsfasman-Vlăduţ-Zink bound the first steps*, IEEE Transactions in Information Theory **43** (Jan., 1997), 128–135.

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